

STUCK

A workbook of research mathematicians' practices at an impasse

Compiled from first-person accounts, with the empirical evidence graded and the exercises done on your own live problem

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Every quotation is printed from a single registry and was matched by machine against a saved text of the source cited beside it. Appendix B gives the record for each quotation.

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How to use this workbook

Being stuck is the ordinary condition of mathematical research. Peter Sarnak says so without softening: “Most of the time one is stuck, and if this is not the case for you, then either you are exceptionally talented or you are tackling problems that you knew how to solve before you started.” This workbook does not try to help you avoid that state. It sets out what research mathematicians say they do while they are in it, and how much of what they say survives contact with the empirical evidence.

The material comes from four kinds of source. Each item is marked with its kind.

Grade	What it means	Examples
T Testimony	A named mathematician’s first-person account of their own practice, quoted exactly from a saved text of the source.	Poincaré (1908), Hadamard (1945), the five “Advice to a Young Mathematician” essays (2008), Tao’s career-advice pages, Fields Medal interviews (2014–2022).
O Observed	A practice recorded in an empirical study of research mathematicians: observed as it happened, or reported in a survey or interview. The sample size and the task are stated.	Savic (2015), nine mathematicians on prepared notes; Liljedahl (2013), 25 respondents; Cranshaw and Kittur (2011), one online collaboration.
L Laboratory	A finding from experimental cognitive science. It is usually from students solving puzzles, with the effect size given and the limits of transfer to research stated.	Sio and Ormerod (2009) and Liu et al. (2026) on incubation; Wagner et al. (2004) on sleep; Thabane et al. (2026) on walking.
C Compiler’s rule	Advice given in this workbook on argument alone, with no source behind it. It is marked so you can tell it apart from the rest.	The two-moves-per-session rule; the stopping rules attached to each move.

The workbook has five parts:

- **Part I** asks you to name the kind of impasse you are in, because “stuck” is not one state and the useful moves differ.
- **Part II** is the catalogue: twenty-three moves. Each gives the practice, the mathematicians’ own words, what the evidence says, an exercise to do on the problem you are actually stuck on, and a rule for when to stop.
- **Part III** supplies the instruments that make the moves repeatable: a stuck log, a session-closing note, a persist-or-abandon contract, a portfolio sheet, and a protocol for testing a move on yourself.
- **Part IV** sets out what the evidence does and does not support, and where the mathematicians flatly disagree with one another.
- **Part V** places the workbook among its predecessors.

Three warnings apply throughout.

1. **The testimony is selected twice.** It comes almost entirely from exceptionally successful people, remembered years later. It is selected by success, and again by the memorability of the moment of insight rather than the months that preceded it.
2. **The laboratory evidence concerns minutes, not months.** It is about students solving word puzzles and number puzzles, and its bearing on a topologist’s eleventh month is an argument, not a measurement.
3. **Nothing here has been tested as a curriculum.** The book is a compilation of what practitioners say and what the science permits, arranged so that you can try it and keep records.

Treat your own log as the experiment, and run it as one: Part III, Instrument I7, says how. None of the exercises uses puzzle problems. Every exercise is to be done on a research problem you are currently stuck on. If you do not have one, put the book down until you do.

Conventions for quotations.

- Every quotation was matched by a script against a saved text of the source cited beside it. For four sources that blocked downloading, the saved text is a text-extraction service's reading of the page, which is weaker; Appendix B names them.
- An ellipsis (...) marks an omission. Square brackets mark words added by this workbook; Weil's translator's own insertions are reproduced as printed.
- Quotation marks inside a quotation are printed as single marks.
- A quotation that stops mid-sentence ends either with an ellipsis or without a full stop inside the closing mark.
- Appendix B gives the match record for every quotation, including the one source whose machine-read text was checked against a scan of the printed page.

Part I. Name the state

The word “stuck” covers several different situations, and the mathematicians’ accounts, read closely, are not all describing the same one.

- **Orientation.** Andrew Wiles’s dark mansion, from the 1997 NOVA documentary on his proof, is an impasse of orientation: “One goes into the first room, and it’s dark, completely dark. One stumbles around bumping into the furniture, and gradually, you learn where each piece of furniture is, and finally, after six months or so, you find the light switch.”
- **Confusion.** William Thurston, on his MathOverflow profile, describes an impasse of confusion that yields to attention rather than search: “Mathematics is a process of staring hard enough with enough perseverance at the fog of muddle and confusion to eventually break through to improved clarity.”
- **Repeated failure.** Terence Tao describes the impasse of an argument that keeps failing at the same place. He handles it by localising the failure rather than looking elsewhere (M3).
- **Failed attempts.** Béla Bollobás describes the full wastepaper basket, which may not be an impasse at all: “If after a session your wastepaper basket is full of notes of failed attempts, you may still be doing very well.”

Experimental work supports the general point that impasse is not one thing. Ross and Arfini (2024) ran four experiments with 856 participants in total, all preregistered, although the first two registered research questions or a design rather than numerical hypotheses. They found that the feeling of being stuck has separable components: felt speed of thought, motivation and affect. Impasses that were later resolved differed from those that were not in felt speed and positive affect, but not in motivation. The practical corollary is that a loss of motivation is not, by itself, evidence that a problem is hopeless. Their impasses lasted minutes, on laboratory problems.

The six states below are this workbook’s own classification (grade C). Ross and Arfini’s components are feelings, not these situations, and nobody has tested whether these six are the right way to divide the ground. Use them as a sorting device.

Uri Alon, writing for biologists, gives the state a picture worth borrowing. In his account of how research actually goes, the path from question to answer meanders: “Experiments stop working, all assumptions seem wrong, and nothing makes sense. The researcher has entered a phase linked with negative emotions that may be called ‘the cloud.’” He treats this as a normal stage of the work rather than a verdict on the researcher, and he adds that “Then, in the midst of confusion, one senses a new problem in the materials at hand.” Mathematicians say the same in their own idiom. Cédric Villani: “My most important discoveries in mathematics, it was by solving some problems that I was not even aware of, say, one or two years before I could solve them.”

Exercise 0. Describe the impasse

Exercise 0 (twenty minutes, in writing)

Write one paragraph describing where you are stuck. Describe the impasse, not the problem:

- What was the last thing that worked?
- What have you tried since, and at what exact point did each attempt fail?
- How long has this been the situation?

Then answer each statement below yes or no. The code in brackets is the state it points to.

- I have no candidate move at all: I do not know what to try next. (S1)
- I have a method, and it fails at the same step every time. (S2)

- I no longer know whether the statement is true. (S3)
- I can follow every line of what I have, and I have lost the picture of why it should work. (S4)
- I have been at this for months and cannot decide whether to abandon it. (S5)
- I cannot make myself sit down with it; the problem has become aversive. (S6)

More than one yes is normal. Use the table that follows to choose where to start in Part II.

The diagnostic table

The right-hand column lists the moves from Part II that the accounts most often recommend for each state; the numbering is that of Part II. The allocation of moves to states is the compiler's reading of the accounts (grade C). Try no more than two moves in one session. That rule is also the compiler's; no source tests it.

State	What it usually looks like	Start with
S1. No candidate move	The page is blank. You reread the statement and nothing suggests itself.	M1 simplest example; M4 dumb questions; M5 change representation; M7 generalise the guess; M12 the overnight question, once M1 has given you computed cases; M15 portfolio
S2. Same failure, same step	A method almost works. The gap is always in the same place.	M3 run the doomed method; M2 hunt the counterexample; M6 do the obvious thing; M8 compute on foot; M23 if a machine is running the method
S3. Truth in doubt	You have stopped believing the conjecture, or believe it only on authority.	M2 hunt the counterexample; M1 simplest example; M9 write it down; M10 say it out loud; M23 if the doubt is about a machine-produced result
S4. Lost the picture	Every line checks, and the whole means nothing. Or a paper you must understand has become a wall.	M5 change representation; M22 three things; M18 let the sea rise; M11 put it down
S5. Persist or abandon	Months in, with sunk cost on one side and guilt on the other.	M17 wastebasket; M21 choose with stuckness in view; M15 portfolio; M19 choose a regime; Instrument I3, the contract
S6. Aversion	You avoid the desk. The problem has become a source of dread.	M16 cap the hours; M11 put it down; M13 walk; M20 accept the state; M14 choose your filler

Two states deserve a warning before you go on.

In **S3**, trying to prove a statement you do not believe is an expensive way to spend a month. Tao's advice is to look for the counterexample first: "if you are trying to prove some ambitious claim, you might try to first look for a counterexample; either you find one, which saves you a lot of time and may well be publishable in its own right, or else you encounter some obstruction, which should give some clue as to what one has to do in order to establish the claim positively ..." This is one mathematician's advice, though the move it recommends is common in the accounts (M2).

In **S6**, none of the desk-level moves will help until the aversion is addressed. The accounts that bear on it concern the shape of the day rather than the shape of the argument: Hardy's and Littlewood's hour limits, and June Huh's three-hour budget and daily walk.

Part II. The moves

Twenty-three practices follow, grouped by the scale at which they operate:

- **At the desk:** within a single working session.
- **Leaving the desk:** across sessions.
- **Across a programme:** over weeks, months or years.
- **When a machine is doing the work:** computation, proof assistants and language models.

Each practice is set out in the same order: the practice itself, the words of the mathematicians who report it, and whatever empirical evidence bears on it. Then comes an exercise to do on your own live problem, and a rule for stopping. The accounts are as clear about the cost of persisting with a move as about its value.

Notice two features of the catalogue before you use it.

The memorable practice is not the commonest. The walk, the shower and waking from sleep dominate the retrospective accounts (Liljedahl, 2013; Sriraman, 2004), because that is where the memorable moment arrived. It is not obviously where the work was done. Only one study has recorded mathematicians' behaviour at an impasse as it happened rather than asking them to remember it (Savic, 2015).

- **The study.** Nine mathematicians worked alone through prepared notes on semigroups, a topic unfamiliar to them, taken from a “proofs course” for beginning graduate students. Six reached an impasse.
- **What they did.** All six moved on to the next item in the notes and came back later; three tried to construct counterexamples, though Savic credits no recovery to one.
- **The caveat.** The notes always supplied a next item, which a research problem does not. Four of the nine, including two of the six who reached an impasse, produced no live recording because of equipment failures, so part of the record comes from exit interviews. The finding is real but bound to its task.

The catalogue contains genuine contradictions. They are set out in Part IV rather than smoothed over here.

At the desk

M1. Start from the simplest example

Scale: desk · *Evidence:* T, O

Before anything else, work the smallest case in which the difficulty is present. That means not simply the smallest case, but the smallest case that already contains the problem. Paul Halmos describes the reflex for Hilbert space, Dusa McDuff describes it for new subjects in general, and Ravi Vakil prescribes it for learning theory. Alan Schoenfeld adds a warning: which special case helps depends on the problem, and the obvious sequence of small cases can mislead.

In their words

“When a mathematician wants to prove a theorem about an infinite-dimensional Hilbert space, he examines its finite-dimensional analogue, he looks in detail at the 2- and 3-dimensional cases, he often tries out a particular numerical case, and he hopes that he will gain thereby an insight that pure definition-juggling has not yielded.”

— Paul Halmos, “Mathematics as a creative art”, *American Scientist* 56 (1968), as reproduced by MacTutor

“When studying something new I often used to start in the middle, using some complicated theory already developed by others. But often one sees further by starting with the simplest questions and examples, because that makes it easier to understand the basic problem and then perhaps to find a new approach to it.”

— Dusa McDuff, “Advice to a Young Mathematician” (2008)

“When you learn the theory, you should try to calculate some toy cases, and think of some explicit basic examples.”

— Ravi Vakil, “For potential Ph.D. students” (Stanford)

What the evidence says

Wilkerson-Jerde and Wilensky (2011) recorded ten mathematicians thinking aloud as they read an unfamiliar research paper on links in geometric topology, with the aim of being able to teach it to a colleague. Eight were professors and two were advanced doctoral students. The experts turned to definitions when questioning or explaining an idea, and to specific instances when making sense of something they did not understand. The task was comprehension rather than an impasse in their own proving, and the sample is ten.

Lockwood, Ellis and Lynch (2016) surveyed mathematicians and interviewed nineteen.

- **Sample.** The survey had 291 respondents, of whom 220 answered the open question the authors analysed. About 55 per cent of respondents were doctoral students and 31 per cent held a doctorate in mathematics.
- **Tasks.** The interviews used conjectures the researchers had set, mostly derived from Putnam competition problems.
- **Finding.** Respondents described building sets of examples that increase in complexity or generality, so that the example works as an instrument rather than a warm-up.

Schoenfeld (1992) gives the warning in concrete form. For one problem, looking at the cases $n = 1, 2, 3, \dots$ suggests the pattern. For a second problem, the same procedure “can lead to a wild goose chase”, and the useful special cases are easily factorable polynomials. “Start from the simplest example” therefore hides a choice: simplest in which respect?

Exercise 1. The three instances

On one sheet, write three instances of your problem.

- **The smallest non-trivial instance:** the one you can compute completely by hand. Compute it completely by hand, and do not stop at “it obviously works”.
- **The smallest instance in which the difficulty is already present.** Identify in writing what makes it different from the first.
- **A boundary or extreme instance:** a degenerate parameter, an infinite family, a case where one hypothesis fails.

Beside each instance, write one line on why you chose it and in which respect it is “simple”. Then answer: does the second instance suggest a statement that is strictly easier than the original and still worth proving? If so, that is your problem for the next session.

When to stop. Stop specialising when the instance you are computing no longer contains the difficulty. At that point you are doing arithmetic, not research, and the move is exhausted. (C)

M2. Hunt for the counterexample

Scale: desk · Evidence: T, O

Spend a bounded amount of time trying to make the statement false. Either you succeed, and

have saved yourself a great deal of effort and possibly gained a result, or the attempt fails at a specific obstruction. Tao's point is that the obstruction tells you what a proof has to deal with. The move has a second use, as a check on an argument that came too easily.

In their words

"if you are trying to prove some ambitious claim, you might try to first look for a counterexample; either you find one, which saves you a lot of time and may well be publishable in its own right, or else you encounter some obstruction, which should give some clue as to what one has to do in order to establish the claim positively ..."

— Terence Tao, "Be sceptical of your own work" (career advice)

"If you unexpectedly find a problem solving itself almost effortlessly, and you can't quite see why, you should try to analyse your solution more sceptically."

— Terence Tao, "Be sceptical of your own work"

What the evidence says

Lockwood, Ellis and Lynch (2016) describe their participants moving back and forth between examples and proof attempts, and between counterexamples and disproof. In Savic (2015), three of the six mathematicians who reached an impasse tried to construct counterexamples, on the set task described above. None recovered that way: one tried unsuccessfully and then took a break, one built a counterexample to a statement he had miscopied, and one made his progress over lunch.

Tao gives the second use a sharper form. An argument that works too easily should be tested for whether "the method may also be able to prove much stronger statements which are known to be false, which would imply that there is a flaw in the method." A quick test follows from this, as the compiler's gloss (C). Remove the key hypothesis and run the argument again. If it still appears to work, it proves too much, and there is an error to find.

Exercise 2. The thirty-minute falsification

Set a timer for thirty minutes and try, in earnest, to make your statement false. Use the boundary instance from Exercise 1, and try to break each hypothesis in turn. Which one, if removed, gives a counterexample, and how quickly?

At the end, write one sentence beginning "The counterexample fails because ...". That sentence names the obstruction. Underline the noun in it; it is often the object the proof needs.

If you have an argument that seems to work, remove its key hypothesis and run it again. If it still "works", it is wrong. Find the line.

When to stop. Stop when the timer stops, unless you have a counterexample in hand. A counterexample search that runs past its budget has become a proof attempt in disguise, and should be logged as one. (C)

M3. Run the doomed method to the point where it fails

Scale: desk · Evidence: T

Apply a technique you already know cannot finish the job, and follow it until it breaks. The point at which it breaks localises the real difficulty far more precisely than thinking about the problem in the abstract. The residue is often small enough to be handled by something crude. This is Tao's doctrine of partial progress, and it turns state S2 (same failure, same step) from a frustration into a measurement. Halmos describes the outcome when the method is wrong in kind.

In their words

“place a high value on partial progress, as being a crucial stepping stone to fully solving the problem”

— Terence Tao, “On the importance of partial progress” (career advice)

“Thus it is not uncommon for months to pass with no visible progress; however by patiently eliminating fruitless avenues of attack, you are setting things up so that when the breakthrough does come, one can conclude the problem in relatively short order.”

— Terence Tao, “Be patient” (career advice)

“It often happens that months of work result in the proof that the method of attack they were based on cannot possibly work, and the process of guessing, visualising, and conclusion-jumping begins again.”

— Paul Halmos, “Mathematics as a creative art” (1968)

What the evidence says

No empirical study isolates this move. The nearest indirect support is Sio and Ormerod’s (2009) meta-analytic finding that a longer period of work before a break goes with a larger benefit from the break. Running a method to its limit is one form of such preparation, but the link is an argument (C), not a finding, and in their data the preparation effect held for visual and creative problems, not for verbal insight problems (M11).

Exercise 3. The failure ledger

Choose the method that has been failing. Run it once more, on paper, from the beginning, without trying to be clever, and number every step. At the step where it fails, stop and write three things:

- (a) the exact inequality, identity or construction that does not hold;
- (b) what additional input would make it hold: a bound, a hypothesis, a lemma;
- (c) whether that input is plausibly true.

Now write down the crudest possible way to supply (b): brute force, a worse constant, an extra assumption. If the crude version works, you have a partial result. Write its statement out in full and date it.

When to stop. Stop when the failure ledger has three entries for the same step and the input required in (b) is the original problem restated. That is circularity. It tells you the method is not merely inadequate but wrong in kind. Move to M5, or to M23 if a machine is running the method. (C)

M4. Ask the dumb questions and answer them

Scale: desk · Evidence: T, O

Interrogate the statement as if you did not trust it:

- Delete a hypothesis and see what breaks.
- Strengthen the conclusion and see whether the proof still goes through.
- Re-prove a standard lemma by a method it is never proved by.

Halmos put it as an instruction for reading, and Tao extends it to research. Both describe the same reflex: turning passive acquaintance with a result into an active understanding of why each part of it is there.

In their words

“Don’t just read it; fight it! Ask your own questions, look for your own examples, discover your own proofs.”

— Paul Halmos, *I Want to Be a Mathematician* (1985), as quoted in the epigraph to Tao’s “Ask yourself dumb questions”

“For instance, given a standard lemma in a subject, you can ask what happens if you delete a hypothesis, or attempt to strengthen the conclusion; if a simple result is usually proven by method X, you can ask whether it can be proven by method Y instead; ...”

— Terence Tao, “Ask yourself dumb questions – and answer them!” (career advice)

“So one should be unafraid to ask ‘stupid’ questions, challenging conventional wisdom on a subject; the answers to these questions will occasionally lead to a surprising conclusion, but more often will simply tell you why the conventional wisdom is there in the first place, which is well worth knowing.”

— Terence Tao, “Ask yourself dumb questions – and answer them!”

What the evidence says

Schoenfeld’s protocol studies are the warrant for the whole family of self-interrogation moves, though they concern students and faculty working unfamiliar problems in a single session.

- **Students.** In his collection of more than a hundred videotapes of students working unfamiliar problems, “roughly sixty percent of the solution attempts are of the ‘read, make a decision quickly, and pursue that direction come hell or high water’ variety.”
- **A mathematician.** One mathematics faculty member working a hard problem spent more than half his time making sense of it, and “as he worked through the problem the mathematician generated enough potential wild goose chases to keep an army of problem solvers busy. But he didn’t get deflected by them.” (Schoenfeld, 1992).
- **The remedy.** Schoenfeld’s course asked students three questions at any moment during group work. He notes that “the better part of a semester is necessary to obtain such changes.” Instrument I2 in Part III is a card version.

The dumb-questions move applies the same interruption to the statement rather than to the solver.

Exercise 4. The hypothesis-deletion table

Write the statement you are trying to prove at the top of a page. Below it, make a table with one row per hypothesis and three columns:

- Delete it: what breaks?
- Weaken it: to what?
- Where in my argument is it used?

Fill every cell. A hypothesis used nowhere in your argument is either unnecessary, in which case a stronger theorem is available, or it is the one your argument is missing, which is why it fails. Either answer is progress.

Then strengthen the conclusion in two different directions. For each, decide in one line whether the strengthened version is plausibly true. A conclusion that cannot be strengthened at all may be sharp, which is worth knowing before you spend a month on it.

When to stop. Stop when every cell is filled. This is a bounded move; it does not need repeating in the same session. (C)

M5. Change the representation

Scale: desk · Evidence: T, O

Stop working in the representation you are stuck in. There are several ways to do this:

- Draw the object, repeatedly, on a large sheet.
- Carry the problem across an analogy into a parallel theory where the same structure is visible from another side.
- Hold several mental models of the same object at once and move between them.

Maryam Mirzakhani drew; Alessio Figalli draws to find the key issues. André Weil described his work as deciphering a text in three languages. William Thurston describes deliberately listening to his own associations, and gives seven ways of thinking about the derivative, from the infinitesimal to the microscopic: “This is a list of different ways of thinking about or conceiving of the derivative, rather than a list of different logical definitions.” He adds: “The list continues; there is no reason for it ever to stop.”

In their words

“But the process of drawing something helps you somehow to stay connected.”

— Maryam Mirzakhani, in Erica Klarreich, *Quanta Magazine* (2014)

“I like drawing pictures. I get an intuition of what’s going on and I can easily catch what are the key issues”

— Alessio Figalli, in Kevin Hartnett, *Quanta Magazine* (2018)

“... my work consists in deciphering a trilingual text {[cf. the Rosetta Stone]}; of each of the three columns I have only disparate fragments ...”

— André Weil, letter to Simone Weil, 26 March 1940 (trans. M. H. Krieger, *Notices of the AMS*, 2005)

“Nothing is more fecund than these slightly adulterous relationships; nothing gives greater pleasure to the connoisseur ...”

— André Weil, letter to Simone Weil, 26 March 1940, on analogies between theories

“I put a lot of effort into ‘listening’ to my intuitions and associations, and building them into metaphors and connections. This involves a kind of simultaneous quieting and focusing of my mind.”

— William Thurston, “On proof and progress in mathematics” (1994)

What the evidence says

Tabatabaeian, O’bi, Landy and Marghetis (2025) filmed PhD-level mathematicians working alone at a blackboard for about an hour each on Putnam competition problems.

- **Sample and coding.** The main analyses cover six mathematicians, twelve sessions and twenty-four spoken insights (“aha”, “ohhhhhh, I see”). With pilot data from a seventh mathematician included, the paper reports fourteen sessions, 4,653 coded shifts of attention between inscriptions and twenty-seven insights, and says the results are unchanged.
- **Finding.** The unpredictability of the shifts (mean surprisal) was higher just before an insight than at other times: 1.23 against 1.07, $t = 2.83$, $P < 0.01$. The rise began more than two minutes before the insight, and surprisal stayed high just after it (1.25).
- **Limits.** The sample is six people on competition problems, and the design is correlational. It is the first evidence that insight in mathematicians coincides with a visible destabilisation

of how they look at the board, which is what a change of representation tries to provoke. It does not show that provoking the destabilisation produces insight.

The often-repeated description of Mirzakhani drawing on huge sheets is not hers. It is her husband Jan Vondrák's, in the same Quanta article: "She has these huge pieces of paper on the floor and spends hours and hours drawing what look to me like the same picture over and over". It is good evidence of the practice, but it is not her own account of it.

Exercise 5. Three columns

Rule a page into three columns, after Weil.

- **Column 1:** write your problem in the representation you have been using.
- **Column 2:** write it in the most different representation you can find that is still faithful: geometric if it was algebraic, combinatorial if it was analytic, a picture if it was a formula.
- **Column 3:** write the representation your nearest neighbouring field would use.

Now list, across the columns, which objects correspond. Where a column has a blank, write the question "what is the analogue of X here?". Those blanks are research questions, and one of them is often easier than your original problem.

If the problem has a geometric or combinatorial side at all, take a sheet larger than A4 and draw the object, then draw it again. Do not annotate.

When to stop. Stop when columns 2 and 3 are merely column 1 in different notation. A representation that changes the symbols without changing what is visible has not been changed. (C)

M6. Do the obvious thing, mechanically

Scale: desk · Evidence: T

When you cannot see the proof, stop trying to see it, and execute the moves that need no insight:

- expand every definition to its lowest level;
- turn every existential quantifier into a named object;
- substitute every named object into every universal hypothesis that accepts it;
- repeat.

Timothy Gowers teaches this as a method for undergraduate analysis, but his point is general. A surprising fraction of the arguments that experienced mathematicians produce "easily" can be generated by a routine, and the routine is available when inspiration is not.

In their words

"the reason experienced mathematicians can come up with certain sorts of arguments in elementary analysis very easily is that these arguments can be generated by a simple algorithm."

— Timothy Gowers, "Solving analysis problems on auto" (dpmms.cam.ac.uk)

"rephrase what you are trying to prove in this lower-level way"

— Timothy Gowers, "How to work out proofs in Analysis I" (blog, 2014)

What the evidence says

There is none at research level. The move is included because it costs little. It is also the routine a proof assistant enforces, which is one reason formalisation sometimes reveals that a “stuck” step was never written out (M23).

Exercise 6. The expansion

1. Take the statement of the step at which you are stuck. Rewrite it with every defined term replaced by its definition, down to quantifiers over elements. Do not summarise; write it all.
2. For every “there exists”, introduce a named object and record its defining property on a separate line.
3. For every “for all” in a hypothesis, list the named objects you now have and substitute each one.
4. Read the resulting list once and mark anything that appears twice. Two occurrences of the same expression, reached by different routes, are often the equation you needed.

When to stop. Stop after one full pass. If the expansion yields nothing, the difficulty is not at the level of definitions and this move will not find it. (C)

M7. Generalise the guess

Scale: desk · *Evidence:* T

Instead of trying to guess the answer, guess its shape with unknown parameters, and let the constraints of the problem fix the parameters. Gowers’s worked example is the cubic formula. Rather than pulling a substitution out of a hat, he asks what made the quadratic case work, and makes a more general guess of the same form. The move cures a particular sterility: staring at a problem and waiting for the right object to occur to you.

In their words

“more general guesses and then work out what their more precise forms must be”

— Timothy Gowers, “Discovering a formula for the cubic” (blog, 2007)

“Could you imagine a more accessible related problem? A more general problem? A more special problem? An analogous problem?”

— George Pólya, *How to Solve It* (1945), “Devising a plan”, as summarised on the one-page list cited

What the evidence says

Pólya’s heuristics have been in print since 1945, but they were never validated as a treatment.

- **Transfer.** Schoenfeld (1992) reports that early studies (Wilson, 1967; Smith, 1973) found taught heuristics did not transfer to new domains.
- **His explanation.** Pólya’s strategies were described rather than prescribed: they let you recognise a strategy when someone else uses it, but not carry it out. His own example is “special cases”, which hides several distinct strategies.
- **Monitoring.** He treats monitoring and control as a further requirement.

Schoenfeld does not report a test of heuristics taught with and without monitoring, so the case for monitoring here is an argument, not a result. The exercise below pairs the guess with a record of why each parameter was introduced, because that record is a form of the monitoring Schoenfeld describes (C).

Exercise 7. The parametrised guess

1. Write down the object you wish existed: a substitution, an auxiliary function, an invariant, a test element. Do not try to specify it.
2. Write its general form with undetermined parameters: a polynomial of unknown degree with unknown coefficients, a function of an unspecified combination of the variables, a sum over an unspecified index set.
3. Impose the constraints of the problem one at a time, and record beside each what it forces on the parameters.
4. Keep going until the parameters are determined or a constraint is contradictory. If a constraint is contradictory, the form was too narrow: add one parameter and repeat. Note which constraint forced the addition; it is the one carrying the information.

When to stop. Stop when the number of free parameters exceeds the number of constraints you can write down. At that point the guess has become a research programme, and belongs in the portfolio (M15), not in this session. (C)

M8. Do the computation in your head, on foot

Scale: desk to session · *Evidence:* T, L

Leave the paper behind and carry the computation. Alain Connes recommends a long walk with no pencil, doing the calculation in one's head whatever the feeling that it is too complicated. Peter Scholze prefers to work without writing anything down, because it forces him to formulate ideas in the cleanest possible way. The practice is not a break from the problem. It is a change of medium that forces compression, since only what can be held in the head survives.

In their words

"One very sane exercise, when fighting with a very complicated problem (often involving computations), is to go for a long walk (no paper or pencil) and do the computation in one's head, irrespective of whether one initially feels that 'it is too complicated to be done like that.'"

— Alain Connes, "Advice to a Young Mathematician" (2008)

"Even if one does not succeed, it trains the live memory and sharpens one's skills."

— Alain Connes, continuing

"You have only some kind of limited capacity in your head, so you can't do too complicated things."

— Peter Scholze, in Erica Klarreich, *Quanta Magazine* (2016)

Quanta Magazine's profile of Hugo Duminil-Copin opens with the journalist's description of him: "With Hugo Duminil-Copin, thinking rarely happens without moving." The words are hers, not his.

What the evidence says

The laboratory evidence on walking concerns generating ideas, not carrying out a computation, so it bears on Connes's claim only indirectly.

The original study. Oppizzo and Schwartz (2014) ran four experiments with 40 to 48 participants each.

- **Divergent tasks.** In Experiment 1, every participant sat and then walked, always in that order. Walking raised scores on the alternate-uses task, a divergent test of generating ideas ($d = 0.70$; 81 per cent of participants improved; about 60 per cent more output on average).

- **Convergent task.** Scores on the compound remote-associates task, a convergent single-answer test, fell ($d = 0.38$).
- **Order.** Because the order was fixed, practice and order are mixed into those figures. Experiment 2 varied the order, and sitting after walking was indistinguishable from walking after sitting at the second test ($p = .975$).
- **Outdoors.** Only Experiment 4 compared outdoor with indoor walking, and on an analogy task the results were mixed.

Later evidence. Thabane et al. (2026) pooled 23 studies from 16 articles (1,036 participants).

- **Divergent tasks.** Walking improved divergent thinking ($d = 0.93$, 95 per cent confidence interval 0.44 to 1.42; 20 studies). Randomised studies alone gave $d = 0.82$ (0.35 to 1.28; 13 studies), which the authors rate as evidence of moderate certainty.
- **Convergent tasks.** It had no reliable effect on convergent thinking ($d = 0.16$, -0.31 to 0.63 ; six studies; certainty rated very low).
- **Indoors versus outdoors.** There was no difference between indoor and outdoor walking.
- **Caveat.** The authors note that many of the positive results come from one author's studies, and call for replication.

Physical activity generally. A wider meta-analysis of physical activity and creativity (Rominger et al., 2022) found smaller effects for single bouts ($g = 0.37$). It found evidence of funnel asymmetry, and it excluded convergent tasks.

A small within-subject study found no effect of a fifteen-minute treadmill walk on four divergent measures (Frith and Loprinzi, 2018; 32 students).

The defensible summary is this. Walking probably helps the generation of options, though the size is uncertain and much of the evidence comes from one laboratory. There is no good evidence that it harms single-answer work: the one decrement rests on a single experiment with a fixed order. A rule to walk for options and sit for answers would go beyond the evidence. Connes's prescription is about compression and attention, for which there is no laboratory test.

Exercise 8. The carried computation

Choose one computation from your current problem that you have been doing on paper and that fits, in principle, on one page. Read it through once. Then walk for at least forty minutes without paper and reconstruct it. When you return, write down:

- (a) where you lost it;
- (b) what you had to simplify in order to carry it.

The simplification in (b) is a candidate lemma: the version of the computation that a head can hold is often the version worth proving.

When to stop. Stop if after three attempts you cannot carry the computation past its first quarter. Compression has failed. That usually means the computation has too many independent parts to be understood as one thing, and needs restructuring on paper first (M9). (C)

M9. Write it down until you can forget it

Scale: desk · *Evidence:* T, O

Write up every argument, including failed ones, to the level of detail at which you could safely forget it and recover it later. Tao does this on a computer, and says it both tests understanding and frees working memory. Halmos, writing about writing mathematics, gives two rules for stopping: write tomorrow's first sentence today, and end on something solid. Littlewood's version is to stop in the middle of something, so that tomorrow begins with an obvious next step. All three aim

at an easy restart. June Huh records everything when the thinking is fast, because his future self will not be able to reconstruct it.

In their words

“write down (preferably on a computer) a sketch of any interesting argument I came across ... detailed enough that I could then safely forget about the details, and readily recover the argument”

— Terence Tao, “Write down what you’ve done” (career advice)

“write tomorrow’s first sentence today; some authors begin today by revising and rewriting the last page or so of yesterday’s work.”

— Paul Halmos, “How to write mathematics” (1970)

“end each work session on an up-beat; give your subconscious something solid to feed on between sessions.”

— Paul Halmos, “How to write mathematics” (1970)

“But try to end in the middle of something; in a job of writing out, stop in the middle of a sentence.”

— J. E. Littlewood, “The mathematician’s art of work” (1967; Mathematical Intelligencer reprint, 1978, p. 117)

“When it’s fast, I try to record as much as possible, because my future self may be unable to recover what I could get in such times. When it’s slow, there is not much I can do about it other than to wait. And I’m pretty good at waiting.”

— June Huh, IMU Fields Medal interview (2022)

What the evidence says

Savic (2015) captured his mathematicians’ writing in real time with tablet computers and smart pens.

- **Time.** Per mathematician, he reports a mean of 2 hours 5 minutes of recorded work (range 1 hour 21 minutes to 2 hours 50 minutes). This was spread over a mean elapsed span of 19 hours 56 minutes, from first start to last stop (range 1 hour 21 minutes to 2 days 5 hours).
- **Pages.** Slightly under 13 written pages each (range 8 to 22).
- **Coverage.** He does not say how many of the nine these averages cover; four produced no live recording.

On any reading, the work was discontinuous, and a written record carried the argument across the gaps. A written note at the close of a session is what gives the interval in M11 something to work on.

Not everyone keeps such a record. Karen Uhlenbeck says she does not save her notes and starts each problem fresh. Andrew Wiles says a slightly bad memory helps (M17). Part IV takes up the disagreement.

Exercise 9. The session-closing note (see also Instrument I4)

From today, finish every session by writing five lines before you stand up:

1. what I believe now that I did not believe this morning;
2. the exact step that is currently blocking;
3. the one thing I would try first tomorrow, written so that it can be started without thought (Halmos’s and Littlewood’s point);
4. one thing that worked, however small (Halmos’s point);
5. the date I will next return.

Store failed arguments in a dated file that is never deleted. Tao keeps even embarrassingly wrong work privately because “you never know whether something salvageable can be extracted from it”.

When to stop. This move has no stopping rule. It is a habit, and Tao and Halmos treat its absence as a cause of lost progress; Uhlenbeck and Wiles do not. If after two months your closing notes are never reread, test the habit on yourself as in Instrument I7. (C)

M10. Say the wrong idea out loud

Scale: desk to session · *Evidence:* T, O

State the idea before it is ready, and in particular before you are confident it is right. Duminil-Copin describes rushing into colleagues’ offices with ideas that were wrong from the first line. Gowers built the Polymath ground rules around the same principle, discouraging participants from going away for a fortnight to work an idea out before posting it. McDuff gives the move its proper scope: interaction matters most while one is learning, and work alone matters once one has the context. Uhlenbeck, who does most of her thinking alone, still finds other people to talk to “absolutely necessary”.

In their words

“I cannot even count the number of times I rushed into the office of one of my postdocs or one of my colleagues to explain some crappy idea that actually was literally wrong from the beginning.”

— Hugo Duminil-Copin, in Jordana Cepelewicz, *Quanta Magazine* (2022)

“When you do research, you are more likely to succeed if you try out lots of stupid ideas. Similarly, stupid comments are welcome here.”

— Timothy Gowers, “Is massively collaborative mathematics possible?” (blog, 2009)

“If you are convinced that you could answer a question, but it would just need a couple of weeks to go away and try a few things out, then still resist the temptation to do that.”

— Timothy Gowers, *Polymath ground rules* (2009)

“These days people are much more aware that mathematics is a communal endeavor: even the most brilliant idea gets meaning only from its relation to the whole. Once one has an understanding of the context, it is often very important and fruitful to work by oneself. However, while one is learning it is vital to interact with others.”

— Dusa McDuff, “Advice to a Young Mathematician” (2008)

“I like to work alone, and I do most of my thinking alone, but I find it absolutely necessary to have other people to talk to now. I need to see how my ideas are panning out.”

— Karen Uhlenbeck, interview with Clara Moskowitz, *Scientific American* (2019)

What the evidence says

Cranshaw and Kittur (2011) analysed the first Polymath project, which set out, in blog comments, to find a combinatorial proof of the density version of the Hales–Jewett theorem.

- **Scale.** There were 1,555 comments on fourteen blog posts between 1 February and 23 May 2009, about sixteen weeks. Of these, 1,228 were the numbered, substantive comments.
- **Participants.** Thirty-nine people wrote at least one numbered comment, of whom 29 could be linked to a real identity.

- **Concentration.** Gowers and Tao together wrote 42 per cent of the numbered comments, and the ten most active participants wrote nearly 90 per cent.
- **The tail.** The participants' own wiki kept a timeline of milestones, which cites 215 numbered comments. These were written by 22 of the 39 commenters (56 per cent). Five people who wrote five or fewer numbered comments in all wrote six of the milestone comments.

"Milestone" here means listed on the participants' timeline, not an independent rating. Read as evidence about impasse, this suggests that the person who resolves a block need not be the person carrying it, which can happen only if the block has been stated where others can see it. That reading is the compiler's (C), and it rests on one project.

Liljedahl (2013) surveyed prominent mathematicians with a questionnaire modelled on Hadamard's. Twenty-five of the 150 approached replied. One of them, Jerry Marsden, placed the moment of insight in conversation: "I can be talking to a colleague or my wife or eating breakfast and suddenly, like a voice from the blue, I get told what to do."

Exercise 10. The written wrong pitch

Write a 200-word statement of your current obstruction, addressed to a named reader who would ask questions rather than solve it. End with your best current idea, and say that you believe it is probably wrong.

Then, on a fresh line, write the three questions that reader would ask first, and answer them. The first of those questions is often the one you had stopped asking.

The accounts above describe telling another person, and the Polymath figures concern other people resolving a block. Writing the pitch captures the act of stating the obstruction, but not the other reader. Whether you show it to anyone is your decision; the exercise does not depend on it.

When to stop. Stop rewriting the same pitch after the third version. Repetition without change means the obstruction is correctly described, and the missing input is not in the description. (C)

Leaving the desk

M11. Put it down, deliberately, with a date to return

Scale: session · *Evidence:* T, O, L

Abandon the problem for a while and take it up again later. The accounts that recommend this include:

- **Hadamard:** his own habit, and the advice he says he always gave beginners.
- **Poincaré:** all three of his famous illuminations.
- **Wiles:** what he says he does when stuck.
- **Uhlenbeck:** her description of an ordinary working day.

Poincaré adds a qualification: the interval is fruitful only if it is preceded and followed by conscious work. On that view, putting down a problem you have not yet worked hard on is not incubation but procrastination with a good name. The laboratory evidence below weakens this for problems that ask for ideas.

In their words

“whether they were in the habit of abandoning a problem for a while to resume it again only later on (which I, personally, do in many cases and which I always recommend to beginners who consult me)”

— Jacques Hadamard, *The Psychology of Invention in the Mathematical Field* (1945), p. 9

“At the moment when I put my foot on the step the idea came to me, without anything in my former thoughts seeming to have paved the way for it, that the transformations I had used to define the Fuchsian functions were identical with those of non-Euclidean geometry.”

— Henri Poincaré, “Mathematical creation” (trans. G. B. Halsted, 1913)

“it is possible, and of a certainty it is only fruitful, if it is on the one hand preceded and on the other hand followed by a period of conscious work.”

— Henri Poincaré, “Mathematical creation”, on unconscious work

“Then you have to stop, let your mind relax a bit and then come back to it. Somehow your subconscious is making connections and you start again, maybe the next afternoon, the next day, the next week even and sometimes it just comes back. Sometimes I put something down for a few months, I come back and it’s obvious.”

— Andrew Wiles, interview, *Plus Magazine* (2016)

“When I’m working on a problem I usually sit down and think through what I know, and make notes and try out a few estimates, do some inequalities. Maybe I go look up a reference. And then I probably get stuck and go on to think about something else.”

— Karen Uhlenbeck, *Scientific American* (2019), describing an ordinary working day

Poincaré adds that “These sudden inspirations (and the examples already cited sufficiently prove this) never happen except after some days of voluntary effort which has appeared absolutely fruitless”.

What the evidence says

The 2009 meta-analysis. Sio and Ormerod (2009) pooled 117 experimental comparisons from 29 publications (3,606 participants; median 25 per study).

- **Overall effect.** The weighted mean effect of an incubation period was $d = 0.29$, with a printed 95 per cent confidence interval of 0.21 to 0.39. The unweighted mean was larger (0.41).
- **Preparation.** Longer preparation before the break went with a larger effect (unstandardised $\beta = 0.03$, $p < .05$). By problem type, this held for visual and creative problems but not for verbal insight problems.
- **Length of the break.** The length of the break itself did not predict the effect.
- **Problem type.** Divergent tasks benefited more than single-answer insight problems in the regression, though the raw subgroup means barely differ.
- **Publication bias.** A test for publication bias was not significant, and no correction was applied.

Since 2009. This compilation found no bias-corrected reanalysis of Sio and Ormerod’s data, but there is a newer synthesis and a preregistered experiment.

- **Liu et al. (2026; published online 2024)** pooled 195 effect sizes from 52 studies of incubation on creative tasks. The mean was $d = 0.39$, 95 per cent confidence interval 0.25 to 0.53, and the effect was moderated by the type of cue given. The abstract reports that the effect

was consistent across cognitive load. It does not say whether publication bias was assessed, and this compilation did not see the full text.

- **McDaniel, Habibi and Kaplan (2025)** ran a preregistered experiment with 200 adults and ten-minute breaks. They found no general benefit of a break over carrying straight on, and no difference between kinds of break (M14). Their samples could detect only moderate or larger effects.
- **Gilhooly et al. (2012)** gave 184 participants a divergent “alternative uses” task. An incubation break taken immediately after the instructions, before any work on the task, did better than one taken after some work. Participants showed no deficit on the task done during the break, which the authors read as evidence for unconscious work rather than for secretly continued conscious work. Note the tension with Poincaré and with the preparation finding above. For divergent tasks at least, a long run of conscious work does not seem to be a precondition.

A caution from a neighbouring literature. In decision-making, the “unconscious thought advantage” fell from a pooled $g = 0.15$ to 0.02 after a standard correction for missing studies. A replication with 399 participants found a Bayes factor of 7.8 in favour of no effect (Nieuwenstein et al., 2015). Incubation in problem solving has not been tested that hard. The fairest reading is a modest laboratory effect, found again in a newer synthesis but not in one preregistered experiment, whose size after any correction for bias is not known.

The observational study. Savic’s own claim about his mathematicians is modest: “For some of the participants’ major incubations described herein, resulting insights occurred, they were helpful, and they moved the argument forward.” On returning to an unfinished proof, “they almost always had a new idea to explore.” Some breaks led nowhere in his own data, and an impasse that was never resolved leaves no recovery to observe.

Exercise 11. The dated abandonment

Do not put the problem down until you have completed the failure ledger (Exercise 3) or the falsification (Exercise 2) in the same session. This rule follows Poincaré and the preparation finding; Gilhooly’s result suggests it matters less for problems that ask for ideas rather than a single answer (C).

Then write on the session-closing note: “I am putting this down until [date], and on that date I will first try [one specific move].” Choose a date at least one night and at most two weeks away. On the return date, reread the failure ledger before you reread the problem.

When to stop. If you have put the same problem down three times in succession and returned with nothing, the problem is not incubating. Move to M15 (portfolio) or Instrument I3 (the contract). (C)

M12. Sleep on it, and know which kind of problem you have

Scale: session · *Evidence:* T, L (contested)

Hadamard reports a long-sought solution appearing at once when he was woken abruptly. Some of Liljedahl’s respondents describe sleep as where the work is done, or the edge of sleep as where ideas arrive. The laboratory evidence supports a narrow version of this claim, and the narrow version has itself held up only weakly.

In their words

“One phenomenon is certain and I can vouch for its absolute certainty: the sudden and immediate appearance of a solution at the very moment of sudden awakening.”

— Jacques Hadamard, *The Psychology of Invention* (1945), p. 8

“On being very abruptly awakened by an external noise, a solution long searched for appeared to me at once without the slightest instant of reflection on my part ...”

— Jacques Hadamard, *The Psychology of Invention* (1945), p. 8

“I’m convinced that I do my best work while asleep. The evidence for this is that I often wake up with the solution to a problem, or at least with a clear idea of how to proceed to solve it.”

— Charles Peskin, respondent in Liljedahl (2013)

Demetrios Christodoulou, another of Liljedahl’s respondents, places his ideas “at night when I am laying in bed, somewhere between consciousness and sleep.” Poincaré gives the one account in this corpus of a sleepless night that worked:

“One evening, contrary to my custom, I drank black coffee and could not sleep. Ideas rose in crowds; I felt them collide until pairs interlocked, so to speak, making a stable combination.”

— Henri Poincaré, “Mathematical creation” (1913)

Hadamard himself separates the experience of sudden awakening, which he vouches for, from the “mathematical dream”, which he says it could easily be confused with.

What the evidence says

The founding study. Wagner, Gais, Haider, Verleger and Born (2004) trained participants on a number-reduction task with a hidden rule which, once noticed, allows a large shortcut.

- **Main experiment.** Three groups of 22 were trained, then spent eight hours in nocturnal sleep, nocturnal wakefulness or daytime wakefulness, and were retested. In the sleep group, 13 of 22 (59.1 per cent) found the rule, against 5 of 22 (22.7 per cent) in each wake group.
- **Without training.** In a supplementary experiment, two groups of 20 were tested without prior training, one after a night’s sleep and one after a day awake. Five of 20 (25 per cent) found the rule in each.
- **Conclusion.** Sleep helped only when it followed training. Of the four comparison groups, three had stayed awake and one had slept without prior training.

Since then. This compilation found no independent direct replication.

- **Vickrey and Lerner (2023).** On the same task with a twelve-hour interval and sleep monitored at home, 4 of 22 participants (18.2 per cent) gained insight after sleep against none of 25 after a day awake. This was one-tailed $p = 0.09$, a trend at a much lower base rate than Wagner’s.
- **Craig, Ottaway and Dewar (2018).** Ten minutes of quiet waking rest after first exposure to the task more than doubled the chance of insight compared with an unrelated perceptual task. On this task, then, the effect is not specific to sleep.
- **Brodt et al. (2018).** For 62 participants, a three-hour break helped classical riddles, and spending the break asleep added nothing.
- **Schönauer et al. (2018).** With 77 participants and a three-hour daytime nap, sleep did nothing for classical insight problems or for working out magic tricks; every sleep-related test had p above 0.4.

- **Lerner and Gluck (2019).** Their systematic review found many null results, and found that the specific task decided whether a sleep effect appeared more than anything else did.

Which stage of sleep. Two nap studies contradict each other, and neither assigned sleep stage at random.

- **Lacaux et al. (2021).** Of 103 participants, 16 found the rule before the nap and were set aside. Among the rest, 20 of 24 who reached only the lightest stage, N1, found the rule, against 15 of 49 who stayed awake and 2 of 14 who reached the deeper stage, N2.
- **Löwe et al. (2025).** On a different, perceptual task, with 68 participants analysed, 24 of 28 who reached N2 had an insight, against 14 of 22 in N1 and 10 of 18 awake. Their preregistered hypotheses, that N1 would beat waking and beat N2, both failed. The N2 advantage over waking ($p = .038$) came from a follow-up analysis.

The fair summary. Sleep after real preparation may help a rule hidden in worked material to surface. But the one strong result has had weak support since, and in a study with no sleep group, quiet waking rest also more than doubled insight on the same task. Sleep has not helped problems that need a representation restructured, and nobody knows which part of sleep, if any, does the work.

In practice. If the impasse is that a pattern in worked examples has not yet declared itself (S1, after M1), the overnight question costs almost nothing and is worth trying. If the whole approach needs reconceiving (S4), the evidence gives no reason to expect sleep to do it, and M5 is the better bet.

Exercise 12. The overnight question

Before sleep, write the question in a form that admits a rule: “Across the cases I computed today, what is the same?” Do not write “How do I prove the theorem?”. List the computed cases beneath it, in a column, in the same notation.

On waking, before anything else, write for five minutes on that page and record what was there. Most mornings it will be nothing; that is expected, and the record is what lets you find out whether it is ever anything.

To make the record informative, run it for four weeks and decide in advance, by the schedule in Instrument I7, which nights get the question and which do not. Write for five minutes every morning either way, and count the mornings with an entry that bears on the problem.

When to stop. If, after four weeks, nights with the question produce no more useful morning entries than nights without, stop keeping the page for that problem. The evidence gives no reason to expect a delayed effect. (C)

M13. Walk, to generate options

Scale: session · *Evidence:* T, L

The walk recurs throughout this corpus:

- Poincaré on the bluff;
- Littlewood’s “relaxed periods ... spent in walking”;
- Kolmogorov’s Sunday walks with his students, out to places thirty kilometres or so away;
- Wendell Fleming crossing campus after a class;
- Hugo Duminil-Copin in the sea;
- Huh’s long walk after lunch each day, reported by Quanta.

The laboratory evidence, set out under M8, says what the walk is probably for: it raises the number and novelty of ideas produced. It does not show any harm to single-answer work.

In their words

“One morning, walking on the bluff, the idea came to me, with just the same characteristics of brevity, suddenness and immediate certainty, that the arithmetic transformations of indeterminate ternary quadratic forms were identical with those of non-Euclidean geometry.”

— Henri Poincaré, “Mathematical creation” (1913)

“There should be relaxed periods during the working day, profitably, I say, spent in walking.”

— J. E. Littlewood, “The mathematician’s art of work” (Intelligencer reprint, p. 116)

“These outings in the environs of Bolshevo, Klyazma, and other places about 30-35 kilometres away, were full of discussions about the current problems of mathematics ...”

— Boris Gnedenko, on Kolmogorov’s Sunday walks, as quoted in the MacTutor biography of Kolmogorov

“This happened while walking across campus after teaching a class.”

— Wendell Fleming, respondent in Liljedahl (2013)

“One day, I was swimming in the sea, and in like 10 minutes of swimming, I had a completely full proof of the other problem,”

— Hugo Duminil-Copin, in Quanta Magazine (2022)

The swim produced the proof of a problem Duminil-Copin was not working on: Quanta’s account is that the idea failed for the question he had been mulling but solved one in a separate field.

Wiles’s walks are easily read as a response to being stuck, but they are his answer to a different question, about how he spends time off:

“I like to go and visit aesthetically beautiful places near Oxford. ... just to go and walk in these landscapes which have been created centuries ago by these people who really devoted their lives to that, I find that very relaxing”

— Andrew Wiles, Plus Magazine, answering “When you take time off – what is your day like?”

What the evidence says

See M8. Walking probably helps idea generation, by an amount that is uncertain and drawn largely from one laboratory (Thabane et al., 2026). Indoor and outdoor walking did not differ. A walk is also a low-demand activity, the kind of break the 2009 meta-analysis rated highest, though that ranking is weak (M14).

Exercise 13. The randomised walk

For four weeks, run a small randomised comparison on yourself (Instrument I7).

1. Before you start, generate a schedule that marks each research day “walk” or “no walk”, balanced in blocks of two days.
2. On walk days, take a walk of at least thirty minutes immediately before your first session, with the problem in mind and no paper.
3. On no-walk days, start the session directly.
4. Either way, begin the session by writing, for twenty minutes, every candidate approach that occurs to you, without evaluating any. Record the count in the log.
5. At the end of the four weeks, run the analysis script.

If you want a measure of quality as well as quantity, have the lists scored later with the day labels hidden.

When to stop. Stop treating the walk as a device for generating options if, after the four-week comparison, walk days do not produce at least one more candidate per session than no-walk days. Keep walking for other reasons if you like. (C)

M14. Choose your filler

Scale: session · *Evidence:* O, L (weak)

What you do during the interval may matter, and the mathematicians and the laboratory point in different directions.

- **The mathematicians.** Observed mathematicians mostly fill an impasse with more mathematics.
- **The 2009 meta-analysis.** Its point estimates favour an undemanding filler over a demanding one.
- **Newer evidence.** It finds little or no effect of filler type at all.

The evidence does not support a strong rule here.

In their words

“I use the calm periods to advance as much as possible the routine work that we get to do as a mathematician: editorial work, refereeing, committees, writing papers, preparing classes, etc.”

— Hugo Duminil-Copin, IMU Fields Medal interview (2022)

Villani: “sometimes also, you know, you get an idea by replying.” Of the questions people send him, he says that sometimes “the effort of writing a reply makes me think”.

June Huh, who often works in the children’s section of a public library, told Quanta: “I don’t like quiet places,” and “It makes me sleepy.” The same profile reports that he likes repetitive, mindless activities such as cleaning and washing up.

What the evidence says

What mathematicians do. Savic’s six mathematicians at an impasse all moved on to the next item in their notes, which the set task always provided. Sriraman (2004) reports that all five of his interviewees worked on several problems at once and switched at dead ends. His own excerpts qualify this: one interviewee says he must think on one thing and not switch so much, and another that he probably works on one, though he moves to another when he gets nowhere. Wiles says he cannot switch (M15).

The 2009 meta-analysis. Sio and Ormerod (2009) give subgroup means of $d = 0.52$ for an undemanding filler (22 studies), 0.46 for rest (16) and 0.24 for a demanding filler (76).

- **Overall.** These differences are not statistically significant.
- **By problem type.** The advantage of an undemanding filler over rest was significant only for verbal insight problems.
- **The demanding filler.** Its penalty appears only in a more complex model with interaction terms.

The three figures look like a ranking; they are not one.

Later evidence.

- **Liu et al. (2026)** report that the incubation effect was consistent across levels of cognitive load.

- **McDaniel, Habibi and Kaplan (2025)** ran a preregistered experiment with 200 adults. They found no difference in creative improvement between a ten-minute easy task, a ten-minute hard task, a meditation exercise and no break. Self-reported mind-wandering during the break predicted improvement when participants returned to the same story prompt, a correlation rather than an effect of the filler. The authors note that their samples could detect only moderate or larger effects.
- **Baird et al. (2012)**. The well-known study that an undemanding task during the break improves later performance had 145 participants, and the benefit was confined to problems seen before the break.
 - *Size*. The figure of about 40 per cent is read off a graph, not printed.
 - *A misprint*. The printed interaction statistic, $F(1, 141) = 4.98, p < .01, \eta^2 = .10$, does not add up. With one numerator degree of freedom $F = 4.98$ gives $p \approx .027$. With three numerator degrees of freedom it gives $p \approx .003$ and $\eta^2 \approx .10$, so the printed degrees of freedom are probably a misprint. That is this compilation's inference, not the authors' statement.
 - *Replication*. A conceptual replication with two studies and 443 participants found no support for the claim that mind-wandering during incubation improves divergent creativity (Murray et al., 2024; abstract only).

Where this leaves the rule. There is a sensible distinction between using an interval to come back to this problem with something new and using it to make progress elsewhere. The evidence does not say which filler serves the first aim best (C).

Exercise 14. Match the filler to the aim

Decide, before you leave the desk, which of two things you want from the interval, and write the letter on the session-closing note:

- (a) to come back to this problem with something new;
- (b) to make progress elsewhere.

If (a), fill the interval with whatever you would choose anyway, keep the failure ledger where you will see it, and do not open another hard problem. If (b), switch fully, and do not expect the first problem to have moved when you return. If you want to know whether the kind of filler matters for you, test it with Instrument I7. The published evidence will not settle it.

When to stop. This is a decision rule, not a repeatable move; apply it each time you leave the desk. (C)

Across a programme

M15. Keep a portfolio of problems

Scale: programme · *Evidence:* T, O (contested)

Keep several problems in hand at once, at different stages and difficulties, and rotate among them. The portfolio turns an impasse from a crisis into a scheduling decision. Its advocates and practitioners include:

- **Sarnak**, who advises it;
- **Rota**, who attributes the method to Feynman, said to keep a dozen favourite problems in mind and to test every new technique against all of them;
- **Hamming**, who counted ten to twenty;
- **Antieau**, describing his own practice before language models.

Two voices qualify it. Villani's advice is sequential rather than concurrent: when stuck, move on

to the next problem. Andrew Wiles says he cannot work this way at all.

In their words

“I often advise students to try to have a range of problems at hand at any given moment.”

— Peter Sarnak, “Advice to a Young Mathematician” (2008)

“You have to keep a dozen of your favorite problems constantly present in your mind”

— Gian-Carlo Rota, “Ten lessons I wish I had been taught” (1997), describing what he calls Feynman’s method

“Most great scientists know many important problems. They have something between 10 and 20 important problems for which they are looking for an attack.”

— Richard Hamming, “You and your research” (1986)

“spend most of your time on more feasible ‘low-hanging fruit’, which will build up your experience, mathematical power, and credibility for when you are ready to tackle the more ambitious projects.”

— Terence Tao, “Don’t prematurely obsess on a single ‘big problem’ or ‘big theory’ ” (career advice)

“Before LLMs, I typically had 5-10 projects on the burners or in the fridge at any one time, probably working actively on 2-3 at any one time.”

— Benjamin Antieau, “Fast math/slow math” (2026)

“If you cannot solve this problem, you will solve the next problem and when you have solved the problem, move to something else.”

— Cédric Villani, interview with Gonit Sora (2017)

“The way some people handle this is they work on several things at once and then they switch from one to another as they get stuck. I can’t do that. I get manic about it. Once I’m stuck on a problem I just can’t think about anything else.”

— Andrew Wiles, *Plus Magazine* (2016)

What the evidence says

- **Sriraman (2004).** He interviewed five research mathematicians at one university, with a protocol adapted from the 1902 survey in *L’Enseignement Mathématique*. He reports that all five worked on several problems concurrently and switched at dead ends; his own excerpts show at least one who tries not to (M14).
- **Johansen and Misfeldt (2014).** They interviewed thirteen research mathematicians at Danish universities, all of whom had been stuck. Several gave the risk of getting stuck as a good reason to work on several problems at once: with a range of things in hand, one is always working and can set aside whatever one is stuck on. One respondent put the base rate plainly:

“Well, by far the greater part of the time you get nowhere. You calculate and you work, and 90 per cent of the ideas you have, if not 95 per cent, lead nowhere.”

— Respondent R10, in Johansen and Misfeldt (2014), *MONA 2014-4*; translated from the Danish by the compiler

- **Alon (2009).** Writing for biologists, he gives the most useful picture. Plot each problem by feasibility and by interest, and look for those on the Pareto front: problems for which no other is both more feasible and more interesting.

Exercise 15. The portfolio sheet (see Instrument I5)

1. List every problem you are currently carrying, including the ones you are ashamed to still be carrying. For each, write one line: what is the next concrete move, and when did you last make one?
2. Draw Alon's axes, with feasibility across and interest up, and place each problem. Circle the front: the problems for which no other problem is both more feasible and more interesting.
3. Make sure the portfolio contains at least one problem you are confident of finishing within a month, and at least one you do not expect to finish this year. This is the compiler's reading of Tao's advice to keep the big problems in view while spending most of your time on feasible ones (C).
4. Rotate to the feasible problem whenever the impasse on the hard one becomes aversive (S6).

If, like Wiles, you find that switching does not work for you, note it and use M19 instead. The accounts do not support imposing a portfolio on someone it does not suit.

When to stop. Prune whenever a problem has gone six months without a concrete move. Move it to the graveyard file (M17), which is not deletion. (C)

M16. Cap the hours, and break them

Scale: programme · *Evidence:* T (partly second-hand)

The testimony from some of the most productive mathematicians on record is about limiting work, not extending it:

- **Hardy**, as reported by C. P. Snow, worked from about nine until one and held four hours of creative work to be the daily limit.
- **Littlewood** recommended four hours a day, or at most five, with a break about every hour, on what his printed text calls days free from research (discussed below).
- **Poincaré**, in the period of his first great discovery, sat at his work table for “an hour or two” a day.
- **June Huh** reports about three hours of focused work a day, a budget that covers preparing lectures and family errands as well as mathematics.

In their words

“Four hours creative work a day is about the limit for a mathematician,”

— C. P. Snow, foreword to Hardy's *A Mathematician's Apology* (1967), reporting Hardy

“On days free from research, and apart from regular holidays, I recommend four hours a day or at most five, with breaks about every hour (for walks perhaps). If you don't have breaks you unconsciously acquire the habit of slowing down.”

— J. E. Littlewood, “The mathematician's art of work” (1967; Intelligencer reprint, 1978, p. 116)

Two sentences later Littlewood turns to teaching: “On days with teaching duties, I can only say, be careful not to overdo the research.” The printed “days free from research” is hard to square with a recommendation of four hours of research. The compiler reads the passage as advice for days without teaching, but that is an interpretation, and the wording is quoted as printed. Elsewhere in the same essay he says: “Either work all out or rest completely.”

“every day I seated myself at my work table, stayed an hour or two, tried a great number of combinations and reached no results.”

— Henri Poincaré, “Mathematical creation” (1913)

“I think intention and willpower ... are highly overrated”

— June Huh, in Jordana Cepelewicz, *Quanta Magazine* (2022); the ellipsis is Quanta’s

Quanta adds: “On any given day, Huh does about three hours of focused work. He might think about a math problem, or prepare to lecture a classroom of students, or schedule doctor’s appointments for his two sons.”

“Mathematicians usually have a hard time explaining to their partner that the times when they work with most intensity are when they are lying down in the dark on a sofa.”

— Alain Connes, “Advice to a Young Mathematician” (2008)

What the evidence says

There is no controlled evidence on daily hours for research mathematicians. The testimony comes from a handful of people whose working conditions were unusual.

Two laboratory findings bear on it only indirectly:

- **Length of the break.** It did not predict the size of the incubation effect in Sio and Ormerod (2009), which removes the argument for long breaks in particular.
- **Some break or none.** An interval generally beats none, which is the case for Littlewood’s hourly break.

Hardy’s figure is Snow’s report, not Hardy’s own text, and it was checked against a third-party copy of the foreword rather than the printed page (Appendix B). Littlewood’s passage was checked against a scan of the 1978 reprint, and is quoted with its opening clause.

Exercise 16. Capped and uncapped weeks

Comparing one week with the next cannot separate the cap from whatever else changed between the weeks, so use Instrument I7.

1. **Plan.** Plan at least four weeks. Pair them, and let the schedule script decide, within each pair, which week is capped (four hours of problem work, a break each hour) and which is uncapped.
2. **Fix the outcome before you start.** A good choice is the number of dated partial results (Exercise 3 statements) written in the week, or the number of closing notes with an entry under “one thing that worked”.
3. **Hide the labels.** Score the notes at the end of the period with the week labels hidden.
4. **Log true hours.** Record the start and stop time of every period of problem work, on paper, in both kinds of week.

Four weeks is very little evidence, and you will know which kind of week you are in. Treat the result as a reason to keep or drop the cap, not as a finding.

When to stop. If capped weeks do not do better on the outcome you fixed in advance, drop the cap. The evidence is testimony, and your log outranks it for you. (C)

M17. Use the wastebasket, and keep what goes into it

Scale: programme · *Evidence:* T (contested)

Abandon dead ends, including large ones, without publishing them, and without deleting them either. Tao’s advice is to store even embarrassingly wrong work privately, because something salvageable can often be extracted later. Bollobás’s full wastepaper basket is the same practice seen from the other side: the failed attempts are evidence of work, not of failure. Wiles disagrees about remembering them, and his reason is worth having.

In their words

“stubbornly working away at a dead end is not the most efficient use of your time, and publishing every last scrap of your work is not always the best way to meet the standards of quality”

— Terence Tao, “Use the wastebasket” (career advice)

“Even an embarrassingly wrong piece of work (and I have a number of these, which fortunately have never made it as far as publication) should be stored privately somewhere, because you never know whether something salvageable can be extracted from it”

— Terence Tao, “Use the wastebasket”

“If after a session your wastepaper basket is full of notes of failed attempts, you may still be doing very well.”

— Béla Bollobás, “Advice to a Young Mathematician” (2008)

“There are two kinds of mistakes. There are fatal mistakes that destroy a theory, but there are also contingent ones, which are useful in testing the stability of a theory.”

— Gian-Carlo Rota, “Ten lessons I wish I had been taught” (1997)

“I really think it’s bad to have too good a memory if you want to be a mathematician. You need a slightly bad memory because you need to forget the way you approached [a problem] the previous time”

— Andrew Wiles, *Plus Magazine* (2016); the square brackets are the magazine’s

“So if you remembered all the failed attempts before, you wouldn’t try them again. But because I have a slightly bad memory I’ll probably try essentially the same thing again and then I realise I was just missing this one little thing I needed to do.”

— Andrew Wiles, continuing

Karen Uhlenbeck: “I actually don’t save my notes; every time I sit down to look at a problem I tend to start thinking it through fresh to see if I find a different way to approach it.”

What the evidence says

There is no evidence either way. The disagreement may be smaller than it looks.

- **Tao** keeps failed work so that pieces can be salvaged.
- **Wiles’s** worry is that remembering a failed attempt as a verdict, that the approach does not work, stops you trying the near-variant that would have worked.

A record of the exact step at which an attempt failed, rather than a verdict on it, serves Tao’s purpose and may meet Wiles’s worry: it shows what nearly worked (C).

Exercise 17. The graveyard file

Create one dated file per abandoned approach, containing:

- the failure ledger from M3;
- the last session-closing note;
- one sentence on why it was abandoned, written as the step that failed, not as a verdict.

Never delete these. Once a month, reread the graveyard for ten minutes. Note any approach whose reason for abandonment no longer holds: a lemma has since been proved, or a tool has since been learned. Note also any approach that failed by one step and might be worth a variant. That review turns Tao’s “salvageable” into a procedure, and it is the place to test Wiles’s point.

When to stop. None. The monthly review is the whole move. If the review never yields anything after six months, reduce it to quarterly. (C)

M18. Let the sea rise

Scale: programme · *Evidence:* T

Instead of striking the problem, surround it. Grothendieck contrasted two ways of opening a nut:

- **The chisel:** strike it until it cracks.
- **Immersion:** submerge it until the shell softens and opens under hand pressure.

His own method was the second: let the general theory rise around the problem until the obstacle dissolves without a blow.

Others give more modest versions:

- **Tao:** if progress requires learning some external piece of mathematics, that is a good thing, not a detour.
- **Atiyah:** follow the tangent when it appears, and cut your losses when it goes nowhere, as he says it does nine times out of ten.
- **Scholze:** he says he learned even linear algebra by assimilating it while learning something else.

In their words

“put the cutting edge of the chisel against the shell and strike hard. If needed, begin again at many different points until the shell cracks”

— Alexander Grothendieck, *Récoltes et Semailles*, on the first method (trans. C. McLarty)

“the sea advances insensibly in silence, nothing seems to happen, nothing moves, the water is so far off you hardly hear it ... yet it finally surrounds the resistant substance”

— Alexander Grothendieck, *Récoltes et Semailles*, pp. 552–553, on the second method (trans. C. McLarty)

“If it turns out that in order to make progress on your problem, you have to learn some external piece of mathematics, this is a good thing”

— Terence Tao, “Don’t be afraid to learn things outside your field” (career advice)

“When this happens to me I always take time out to pursue the idea to see where it leads or whether it will stand up to scrutiny. Nine times out of ten it turns out to be a blind alley, but occasionally one strikes gold. The difficulty is in knowing when an idea that is initially promising is in fact going nowhere. At this stage one has to cut one’s losses and return to the main road.”

— Michael Atiyah, “Advice to a Young Mathematician” (2008), on a question that arises in passing

“I never really learned the basic things like linear algebra, actually — I only assimilated it through learning some other stuff.”

— Peter Scholze, in *Quanta Magazine* (2016)

What the evidence says

There is none at research level, and the practice is the one most obviously confounded with talent: Grothendieck's sea rose because he could raise it. The compiler's view is that the move suits state S4 (lost the picture) and does not suit S2 (same failure, same step), where localising the failure (M3) is cheaper and faster (C).

Exercise 18. The theory that would make it trivial

Write the sentence "This would be trivial if I had a theory of ..." and complete it three different ways. For each completion, write:

- whether such a theory exists;
- whether it exists in a neighbouring field under another name;
- what its first non-trivial example would be.

Pick the completion that exists elsewhere. Spend the next fortnight of problem time learning its first chapter, with your problem as the running example. Do not work on the problem directly during the fortnight.

When to stop. Stop after the fortnight and return to the problem with M1. If the new theory has not changed which instance you consider simplest, it was the wrong theory to learn. (C)

M19. Bear it in mind all the time, or do not: choose

Scale: programme · *Evidence:* T (contradictory)

Here the accounts flatly disagree, and the disagreement is worth having in front of you rather than resolving.

- **Immersion.** Bollobás holds that the most useful device is to keep the problem in mind all the time, as Newton did. Hamming's mechanism is saturation: if you are immersed day after day, the unconscious has nothing else to work on. Littlewood says that after a month's immersion the mind knows its problem intimately. Wiles, stuck, cannot think about anything else (M15).
- **Disengagement.** Hadamard abandoned problems deliberately and told beginners to do the same (M11). Poincaré's illuminations came after travel had made him forget his work, or after he had gone to the seaside, disgusted with his failure, and thought of something else.

Both camps appeal to unconscious work, and they prescribe opposite regimes for it.

In their words

"When thinking about a problem, perhaps the most useful device you can employ is to bear the problem in mind all the time: it worked for Newton, and it has worked for many a mortal as well."

— Béla Bollobás, "Advice to a Young Mathematician" (2008)

"If you are deeply immersed and committed to a topic, day after day after day, your subconscious has nothing to do but work on your problem."

— Richard Hamming, "You and your research" (1986)

"after a month's immersion in research the mind knows its problem much as one's tongue knows the inside of one's mouth."

— J. E. Littlewood, "The mathematician's art of work" (Intelligencer reprint, p. 116)

“Most striking at first is this appearance of sudden illumination, a manifest sign of long, unconscious prior work.”

— Henri Poincaré, “Mathematical creation” (1913)

Of the Coutances journey, Poincaré writes: “The changes of travel made me forget my mathematical work.” Of the later discovery: “Disgusted with my failure, I went to spend a few days at the seaside, and thought of something else.”

What the evidence says

The laboratory cannot settle this at the relevant timescale; its breaks last minutes, or at most a night. What it offers is mixed:

- **Preparation.** Some prior work matters for some kinds of problem (Sio and Ormerod, 2009).
- **Immediate breaks.** For tasks that ask for ideas, a break taken at once can work (Gilhooly et al., 2012).
- **Length of the break.** It does not predict the effect.
- **Motivation.** Ross and Arfini (2024) add that motivation does not distinguish impasses that are later resolved from those that are not. So the felt urgency of keeping a problem always in mind is not, by itself, evidence that the regime is working.

Exercise 19. Alternating regimes

A single fortnight of each regime cannot separate the regime from the passage of time, so use Instrument I7.

1. **Blocks.** Plan at least four blocks of two weeks on the same problem: two of immersion and two of abandonment, with the order within each pair decided by the schedule script.
 - *Immersion:* the problem on the wall, on the walk, at breakfast; no other research problem.
 - *Abandonment:* bounded sessions, dated abandonments (M11), and the problem out of sight between sessions.
2. **Before you start,** fix two outcomes: the number of closing-note entries under “one thing that worked”, scored blind at the end; and the number of days on which you did not want to sit down. The second is the cost, and it is real.

Eight weeks on one problem is a long test with very few units. Treat the result as a choice of regime for that problem.

When to stop. Choose one regime for each problem, not one for life. Most of the mathematicians who describe both regimes are describing different problems. (C)

M20. Accept the state

Scale: programme · *Evidence:* T, L

Wiles’s claim is not that professional mathematicians get stuck less than others, but that they have learned to bear it. Others describe the same disposition:

- **Thurston** treats confusion as the working medium.
- **Mirzakhani** said she did not get easily disappointed.
- **Huh** says he is good at waiting.
- **Tao** says there are remarkably few “Eureka!” moments.

This is not consolation. It is a claim that the drive to keep going is separable from what the impasse tells you about the problem. Recent experimental work is consistent with that, though it also finds that some feelings did carry information.

In their words

“What you have to handle when you start doing mathematics as an older child or as an adult is accepting the state of being stuck ... People don’t get used to that. They find it very stressful.”

— Andrew Wiles, Heidelberg Laureate Forum press conference (2016), as transcribed by Ben Orlin (2017); the ellipsis replaces Orlin’s “Wiles said”

“It’s part of the process. It’s not something to be frightened of.”

— Andrew Wiles, the same occasion

“We all find it difficult. It’s not that we’re any different from someone who struggles with maths problems in third grade.... We’re just prepared to handle that struggle on a much larger scale. We’ve built up resistance to those setbacks.”

— Andrew Wiles, Heidelberg Laureate Forum (2016), as transcribed by Ben Orlin; the ellipsis is Orlin’s

“I’m happy when I can admit, at least to myself, that my thinking is muddled, and I try to overcome the embarrassment that I might reveal ignorance or confusion.”

— William Thurston, MathOverflow user profile, as reproduced by Cornell University

Maryam Mirzakhani told Quanta: “I don’t get easily disappointed,” and “I’m quite confident, in some sense.”

“Incidentally, most problems are solved primarily by this sort of patient, thoughtful attack; there are remarkably few ‘Eureka!’ moments in this business, and don’t be discouraged if they don’t magically appear for you (they certainly don’t for me).”

— Terence Tao, “Be patient” (career advice)

What the evidence says

The laboratory. Ross and Arfini (2024) ran four preregistered experiments with 856 participants. Impasses that were later resolved differed from those that were not in felt speed of thought and positive affect, but not in motivation. A feeling of insight could occur without any preceding impasse. The problems were solved in minutes, and the transfer to months is an assumption. But the finding that motivation is the same across resolved and unresolved impasses is what Wiles describes from the other side: the drive to keep going is not what distinguishes the problems that yield.

The survey. Liljedahl (2013) sent Hadamard-style questions to 150 prominent mathematicians and received replies from 25.

- **Prior work.** All the respondents described prolonged conscious work as important.
- **Intensity.** In Liljedahl’s words, “some of the mathematicians go further to point out that the more intense this period of conscious work is the more intense the AHA! is.” It is “some”, not all, and he quotes two respondents.

This is the kind of retrospective claim that selective memory would produce. Hold it lightly.

Exercise 20. The reread, with a comparison

1. In the stuck log (Instrument I1), mark every entry in which you wrote that the problem was hopeless; the log has a column for this.
2. Once a month, run the analysis script, or count by hand. Find the proportion of “hopeless” entries followed within thirty days by an entry recording progress on the same problem.

3. Find the same proportion for entries that were not marked hopeless.

Write both proportions on the front of the log. The first number cannot tell you anything on its own. If the two proportions are close, the feeling of hopelessness carries no information about your problems; if the first is much lower, it does.

When to stop. None. The state does not end; the relation to it changes. (C)

M21. Choose problems with the risk of stuckness in view

Scale: programme · Evidence: T, O

The most effective response to impasse may happen before any work begins, in the choice of problem.

- **Misfeldt and Johansen** found that research mathematicians weigh the risk of getting stuck when choosing what to work on.
- **Tao's** rule is to aim at the easiest problems of interest that current tools cannot quite handle, and to treat neglect of everything else for one big problem as a warning sign.
- **Alon's** rule for his laboratory is to spend three months choosing before committing.

In their words

"the easiest problems of interest that you can't quite completely handle with your existing tools"

— Terence Tao, "Continually aim just beyond your current range" (career advice)

"When one begins to neglect other tasks (such as writing and publishing one's 'lesser' results), hoping to use the eventual 'big payoff' of solving a major problem or establishing a revolutionary new theory to compensate for lack of progress in all other areas of one's career, then this is a strong warning sign that one should rebalance one's priorities."

— Terence Tao, "Don't prematurely obsess on a single 'big problem' or 'big theory' "

"And never be too sure of what you want to do."

— Cédric Villani, interview with Gonit Sora (2017)

"Mathematical research is by its nature unpredictable – if we knew in advance what the answer would be and how to do it, it wouldn't be research!"

— Terence Tao, "Be flexible" (career advice)

Uri Alon's rule for new students and postdoctoral researchers in his laboratory: "Do not commit to a problem before 3 months have elapsed."

What the evidence says

Misfeldt and Johansen (2015). They interviewed research mathematicians at Danish universities and identified four criteria for choosing a problem: personal interest, continuity with previous work, the danger of getting stuck, and the expected response of colleagues.

- **Sample.** The 2015 paper's abstract does not give the sample size. Johansen and Misfeldt's companion paper (2014), which reports the same interview study in Danish, gives thirteen semi-structured interviews.
- **Interview topics.** Problem choice came up spontaneously in the first four and was asked about directly in the remaining nine.
- **Being stuck.** All thirteen had been stuck, and several said that the risk of it was a reason to keep several problems going.

Alon (2009) writes for experimental biologists, and the transfer to mathematics is by analogy.

Exercise 21. The pre-mortem

For the next problem you take up, write before the first session one paragraph beginning “Six months from now this has failed, and the reason was ...”. Write three such paragraphs, with three different reasons. For each reason, write:

- the earliest observable sign that it is happening;
- the move from Part II you will make when you see it.

File this with the problem’s first session-closing note, and reread it at the first serious impasse.

When to stop. Write a pre-mortem once per problem. Do not revise it later to make it look prescient. (C)

M22. Three things

Scale: programme · Evidence: T, O

Some impasses are of input rather than output: state S4, when the wall is a paper or a seminar rather than your own argument. For these, Ravi Vakil prescribes a discipline. Leave every talk with exactly three things written down, each concrete enough to be a definition, a theorem, an example or a question. When a fourth arrives, one of the three must be cut. Vakil is candid that the exercise is a little contrived. Karen Uhlenbeck reads a paper by looking at the beginning, the end, the references, the main theorems, the definitions and the key lemmas, then trying to prove the key lemmas herself and going back to the paper only when stuck.

In their words

“Your goal is to have three things, and only three things, on this sheet at the end of the talk.”

— Ravi Vakil, “The ‘three things’ exercise for getting things out of talks” (Stanford)

“Something vague (‘I liked the part where she talked about groups’) does not count as a ‘thing’.”

— Ravi Vakil, “The ‘three things’ exercise”

“It is admittedly a little contrived, and when a bunch of us first experimented with it (perhaps around 2007?), we stopped doing it after a while because we got tired of it.”

— Ravi Vakil, *ibid.*

Of the questions that survive, Vakil writes: “Don’t let them drop.”

“I look at the beginning, I look at the end, I look at the references, I try to find the main theorems, I try to find the definitions, and then I try to find the key lemmas. Then I try to prove the key lemmas, and when I get stuck on the key lemmas I go back and look at the paper.”

— Karen Uhlenbeck, interview with Bjørn Ian Dundas and Christian Skau (EMS Newsletter, 2019)

“The very best way to read a book, however, with, to be sure, pencil and paper on the side, is to keep the pencil busy on the paper and throw the book away.”

— Paul Halmos, “The problem of learning to teach”, *American Mathematical Monthly* 82 (1975), as reprinted in *Notices of the AMS* (2007)

What the evidence says

Wilkerson-Jerde and Wilensky (2011), described under M1, watched ten mathematicians read an unfamiliar research paper. When the experts did not understand something, they turned to examples and other specific instances rather than reading on. The three-things exercise imposes a similar pause on a talk, where the pace does not allow one to happen naturally (C).

Exercise 22. Three things, applied to the paper you are stuck on

1. Take the paper or chapter you cannot get through. Read only its statement of results, its final section and its reference list, and write three things.
2. Find the definitions and the key lemmas, and try to prove the lemmas yourself for one hour without reading their proofs.
3. Open the proofs only at the point where your attempt failed. Read from there until the failure is repaired, then close it and continue your own attempt. Repeat.
4. At the end, rewrite the three things. If none of them changed, you have not yet understood the paper; if all three changed, you have.

When to stop. Stop reading when the three things are stable across two sessions. Further reading is then reference, not understanding. (C)

When a machine is doing the work

M23. Stop scaling, check the target, replay the result

Scale: desk to programme · *Evidence:* T, O (recent, small, partly from tool developers)

Computation, search, proof assistants and language models change the shape of an impasse more than they remove it. The accounts in this section, most of them from 2021 to 2026, describe three ways a machine-assisted impasse goes wrong and several ways the machine helps.

The three failures are:

- **Escalation:** more runs, larger parameters, more computing power, on a method that is wrong in kind. This is M3's circularity, paid for in computing time.
- **The wrong target:** the machine optimises the measure you gave it rather than the problem you meant.
- **Accepting too early:** trusting an output that came too easily, or one that answers a nearby problem that was already solved.

The help is of three kinds: quick falsification, small cases from which a person can see the pattern, and holding an argument too large for one head.

The tools change faster than any workbook. Read the specific failures below as examples of a pattern, not as statements about current systems.

In their words: escalation

"Thus the motto of a book I published in 1961: The purpose of computing is insight, not numbers."

— Richard Hamming, *The Art of Doing Science and Engineering* (1997), p. 21

"But that is why I am suspicious, to this day, of getting too many solutions and not doing enough very careful thinking about what you have seen. Volume output seems to me to be a poor substitute for acquiring an intimate feeling for the situation being simulated."

— Richard Hamming, *The Art of Doing Science and Engineering* (1997), p. 131

“The results highlighted our inexperience. Since we forced our own heuristic search method (trying the predefined set of simple transformations) onto AlphaEvolve, it was much more restricted and did not do well.”

— Bogdan Georgiev, Javier Gómez-Serrano, Terence Tao and Adam Zsolt Wagner, “Mathematical exploration and discovery at scale” (2025), §6

The same authors conclude that “for problems where genuinely new, deep insights are required to make progress, AlphaEvolve is likely not the right tool to use.”

“I can attest that there is a tendency to do more and more and more, but that the pleasure in doing so drops, the quality is hard to verify, and my understanding of the output goes down as well.”

— Benjamin Antieau, “Fast math/slow math” (blog, 2026)

Antieau is reporting seven months of near-continuous use of agentic coding systems, in a post whose “golden rule” is: “Do not work faster than you can understand or effectively communicate.”

In their words: the wrong target

The same AlphaEvolve episode continues. Once the authors let the system propose arbitrary functions:

“it always eventually figured out a way to cheat by suggesting a highly irregular function that exploited the numerical integration methods in our scoring function in just the right way, and got impossibly high scores.”

— Georgiev, Gómez-Serrano, Tao and Wagner (2025), §6

A participant in a study of mathematicians using the same system said that when they “thought it was doing well, it was cheating.” Tao, on computer-assisted proof in general:

“Finally, even if the code executes perfectly, the expression that is correctly computed by the code may not be the expression that one actually wanted for the mathematical argument.”

— Terence Tao, “Machine-assisted proof”, Notices of the AMS (2025), accepted manuscript

In their words: accepting too early

“Another common response given by LLM’s was to change the problem to a related but different, and already-solved problem”

— Lauren Williams, in Abouzaid et al., “First proof” (2026), Appendix A.3

“After five minutes, it gave me something confident, plausible-looking, and (I could tell) wrong. But rather than laughing at the silly AI like a skeptic might do, I told GPT5 how I knew it was wrong.”

— Scott Aaronson, “The QMA singularity” (Shtetl-Optimized, 2025)

“Essentially the model proposed the same strategy that was already identified in the most recent work on the problem (and which I restated in the blog post), but did not offer any creative variants of that strategy.”

— Terence Tao, Mastodon post, 15 September 2024

“it has a tendency to ignore facts that do not support its main conclusion (even if it itself has generated those facts), and to invent facts that do support it”

— Expert case study by Timothy Gowers, in Collins et al., PNAS (2024), on GPT-4

Aaronson's episode ended well, because he knew why the answer was wrong and could say so. His summary of what such a model offers: "it can help you get unstuck if you otherwise know what you're doing, which you might call a sweet spot."

In their words: what the machine is good for

"Using computer tools to test conjectures is by itself a major timesaver for mathematicians, as it permits them to quickly rule out false notions."

— David Bailey and Jonathan Borwein, "Experimental mathematics: Examples, methods and implications", *Notices of the AMS* (2005), p. 502

"I could not immediately see the pattern here, but after some trial and error in which I tried to align numerators and denominators, I eventually organized this sequence into a more suggestive form"

— Terence Tao, "The story of Erdős problem #1026" (blog, 2025)

The context is Tao's account of Erdős problem #1026. An hour of AlphaEvolve runs produced numerical bounds for small cases. Tao turned the numbers into exact fractions, saw the pattern by hand, and so arrived at a conjecture that collaborators then settled.

"doing simple simulations at the early stages lets you get insights into the whole system which would be disguised in any full scale simulation."

— Richard Hamming, *The Art of Doing Science and Engineering* (1997), p. 131

"Really, one key problem I had when I was trying to find this proof was that I was essentially unable to keep all the objects in my 'RAM', and I think the same problem occurs when trying to read the proof."

— Peter Scholze, "Half a year of the Liquid Tensor Experiment" (Xena blog, 2021)

Scholze is describing the Lean formalisation of his work with Clausen, which others carried out while he followed and answered questions. The proof assistant shows the current goal at every step, which helped with an argument too large to hold in mind. He also records the limit:

"So I have gained the reassurance that it is possible to climb the mountain along this route, but I still have no sense of the terrain."

— Peter Scholze, "Half a year of the Liquid Tensor Experiment" (2021)

"every time I've seen Lean not accept a human argument in the mathematical literature, the human has been wrong."

— Kevin Buzzard, "Can AI do maths yet?" (Xena blog, 2024)

What the evidence says

All of the studies are recent and small. Two were written by the developers of the tools they study, and four of the five have not been peer reviewed.

Collins et al. (2024). Volunteers tried to prove undergraduate-level theorems with help from language models of 2023 vintage, and rated every response. Three models were used, and participants did not know which they had. The data comprise 25 entries and 261 interactions.

- **Correctness and helpfulness.** Ratings of the two were highly correlated ($r = 0.83$), but some incorrect answers were rated helpful.
- **Misplaced trust.** Some participants who lacked confidence in solving the problem themselves rated a generation completely correct when it was not.
- **The authors' advice:** "Be careful not to fall into the trap of lazy checking".

Feng et al. (2026; preprint; Google DeepMind authors). They ran a language-model agent against the 700 Erdős problems then listed as open, and had mathematicians grade the 212 answers its own verifier flagged.

- **Grades.** Of 200 answers definitively graded, 137 (68.5 per cent) were fundamentally flawed and 63 (31.5 per cent) technically correct; only 13 (6.5 per cent) were meaningfully correct.
- **Misread problems.** The other 50 correct answers were technically valid but vacuous, because they had misread what Erdős intended.
- **Novelty.** Of the 13, nine were already in the literature.
- **Cost of checking.** The authors write that “the lengthiest and most arduous step of our effort was the final one of investigating whether the solutions were already in the literature, and whether they really addressed the intended problem.”

Bäuerle et al. (2026; preprint; Google DeepMind authors). Eleven mathematicians used AlphaEvolve for three months and ran more than 2,300 experiments; they found new results and published papers from them.

- **Reward hacking.** It was common, even with an agent built to critique the outputs.
- **Loops.** The authors note that “the LLM-driven process can get caught in non-productive loops, perhaps incrementally improving on the score by virtue of random chance without making real progress.”
- **Deciding when to stop.** Users judged whether to continue a run by watching how its score moved.

Collins et al. (2026; preprint). This study looked at writing formal proofs in the Lean proof assistant.

- **Survey.** Of 31 respondents, the commonest frustration with AI tools was unreliability.
- **User study.** Seven participants formalised problems with and without AI tools, 42 trials in all. Tool access raised the probability of a fully correct formalisation by 0.30 (95 per cent confidence interval 0.10 to 0.50), with no detectable change in time taken.

Bolan et al. (2025). This is the report of the Equational Theories Project, a large collaboration of people and automated provers. On the hardest implications, which resisted automated approaches, the participants found that “in most of the difficult implications that resisted automated approaches, we found that LLMs did not provide useful suggestions beyond what the human participants could already propose.”

Summary. Nothing here measures what happens to a research mathematician’s impasse when a machine is added. The studies agree on three practical points:

- unchecked outputs are often wrong, or right about the wrong problem;
- systems given a score will find ways to raise it that do not serve the problem;
- checking, including the search for prior work, is a large part of the cost.

Exercise 23. The escalation ledger

Before any run larger than the last, write two lines:

1. What result would show that the method is wrong in kind, not merely short of power?
2. What did the previous scale-up change at the failing step? If two successive scale-ups have left the failing step unchanged, stop: you are in M3’s circularity, and more computing will not get you out.

For every machine-produced result you intend to rely on, write four more:

3. **The target.** What was the machine optimising or checking, and how could it score well without solving your problem? Test that route once.

4. **The problem answered.** Write your problem and the problem the output answers side by side. If they differ, the output answers a nearby problem.
5. **The replay.** Reproduce the result by an independent route before it enters a closing note as fact: a different program, exact arithmetic, a proof assistant, or a hand check of a small case.
6. **The literature.** Search for the result before treating it as new.

When to stop. Stop scaling when two successive scale-ups leave the failing step unchanged, and return to M3 and M5. Stop trusting a result that has not passed lines 3 to 6. (C)

Part III. Instruments

The moves in Part II are only as good as the records they leave. Mason, Burton and Stacey's *Thinking Mathematically* (1982; 2nd edn 2010) is a well-known book with chapters on the stuck state: "Responses to being STUCK" and "Still STUCK?". It asks readers to keep a written record of their work in a set format. It was designed for problems solved in an afternoon. The instruments below adapt the idea to the months-long timescale of research, and add the decision rules that research needs and the afternoon problem does not.

The bundle that accompanies this workbook contains a spreadsheet template for the stuck log and two short scripts: one generates randomised schedules for Instrument I7, and the other analyses the log. Paper works as well.

I1. The stuck log

Keep one line per session, for the life of the problem. The state code is from Part I and the move code from Part II.

Column	What to write
Date	
Problem	Short name, as on the portfolio sheet (I5)
State	S1–S6
Move	M1–M23
Exact failing step	The precise step at which this session's attempt failed. "Didn't work" is not an entry.
What came to mind	However small
Hopeless?	Yes if, at the time, you felt the problem was hopeless (used in Exercise 20)
Progress?	Yes if this session produced something you would put under "one thing that worked"
Condition	Only during a self-experiment (I7): the condition the schedule assigned
Outcome	Only during a self-experiment: the measure fixed in advance
Return date	

The column most often left blank, and the one that matters most, is the exact failing step. It is what lets the graveyard review (M17) find an approach that failed by one step.

Margin marks in the working itself, written as they happen, make the log easier to fill in:

- **STUCK**, with the time;
- **AHA**, with what came to mind;
- **CHECK**, when a claim is checked against an instance;
- **REFLECT**, with what the last hour taught, in one sentence.

Writing STUCK at the moment it occurs means the log records how long the state lasted, which no one remembers accurately afterwards. These marks follow the spirit of Mason, Burton and Stacey's written record. Their book's exact keyword set could not be checked (Appendix B), so the set here is this workbook's own (C).

12. The Schoenfeld card

Alan Schoenfeld's studies of problem solvers found that the decisive difference between novices and experts was whether they stopped to ask if the current approach was working. Roughly sixty per cent of the student attempts in his videotapes were, in his phrase, of the "read, make a decision quickly, and pursue that direction come hell or high water" variety. In his problem-solving course, he reserved the right to ask three questions of any group at any time (Schoenfeld, 1992). Write them on a card and prop it against the screen. Ask them every twenty minutes, aloud if alone.

The card

"What (exactly) are you doing? (Can you describe it precisely?)"

"Why are you doing it? (How does it fit into the solution?)"

"How does it help you? (What will you do with the outcome when you obtain it?)"

If you cannot name what you are doing, you are not doing a move. If nothing will be different on the page in twenty minutes when it works, stop.

Schoenfeld reports that it took his students the better part of a semester for the questions to become habitual. Expect the card to feel artificial for some weeks.

13. The persist-or-abandon contract

State S5 is the one the accounts handle worst, because the people giving the accounts are those who persisted and were right. Tao counsels flexibility; Bollobás says to keep the problem in mind all the time; both are advice from the far side of the decision. The contract below makes the decision in advance, when you are not yet stuck. That is the only time it can be made without the pull of sunk cost (C).

Contract (complete at the start of a problem; do not amend)

- Problem: _____ Date started: _____
- I will regard this problem as having stalled if, by _____ (date), I have not achieved: _____ (a concrete partial result, not "progress").
- If it has stalled, I will do the following before deciding:
 - (a) run M3 and file the ledger;
 - (b) write the wrong pitch (M10);
 - (c) if a machine has been doing much of the work, complete the escalation ledger (M23);
 - (d) reread the graveyard file (M17) for this problem.
- If after (a)–(d) the ledger's required input is still the original problem restated, I will move the problem to the portfolio's reserve list and not touch it for _____ months. This is not abandonment; the graveyard is kept.
- Signed: _____

14. The session-closing note

Write five lines every session, before standing up. Halmos's rule was to end on something solid; Littlewood's was to stop in the middle of something; Tao's was to write until the details can safely be forgotten. The note is what makes a dated abandonment (M11) different from simply stopping.

Session-closing note

1. What I believe now that I did not believe at the start of the session:
2. The exact step currently blocking:
3. The one thing I will try first next time (written so it can be started without thought):
4. One thing that worked, however small:
5. Return date, and the aim for the interval (a: return with something new; b: progress elsewhere):

I5. The portfolio sheet

This sheet follows Sarnak, Rota and Hamming, and uses Alon's axes. Redraw it quarterly. A problem that has not moved between two redrawings is a candidate for the reserve list.

Problem	Feasibility (1–5)	Interest (1–5)	Last concrete move (date)	Next concrete move	Status (active / reserve / graveyard)

The rules for the sheet are the compiler's reading of the accounts (C):

- Keep at least one problem you expect to finish within a month, and one you do not expect to finish this year.
- Keep the total between a dozen and twenty if you can (Rota, Hamming), and fewer if you cannot hold them in mind; the method depends on testing new tools against all of them.
- Do not let any single problem become the one whose payoff justifies neglecting the others (Tao's warning sign).
- If, like Wiles, you cannot switch, keep the sheet for choosing the next problem, not for rotating among current ones.

I6. The weekly review

Spend fifteen minutes at the end of the week, with the stuck log, the closing notes and the portfolio sheet open.

1. Count the sessions and the true hours (M16). Note the longest unbroken period.
2. For each session, check that the exact-failing-step column is filled. Fill any blank from memory now, before it is gone.
3. Count the sessions that ended with an entry under "one thing that worked". If fewer than half did, the problem may be in S6 territory, and next week should be shaped by M16, M11 and M13 rather than by the desk moves (C).
4. Check the return dates. A dated abandonment whose date has passed without a return is a decision, not an oversight; record which.
5. Choose next week's first move, and write it at the top of a fresh page.

I7. Testing a move on yourself

Comparing one week, or one fortnight, with the next cannot separate a move from everything else that changed in the meantime: the problem, your mood, the term, the weather. Here is a protocol that can, within the limits of a single person (C).

1. **One question.** Choose one move and one alternative: walk or no walk before the first session; an overnight question or none; capped or uncapped weeks; immersion or abandonment.
2. **The unit.** Use the day for moves that act within a day, and blocks of one or two weeks for regimes. Many short units beat a few long ones.
3. **Fix the outcome in advance.** Write it down before the first unit. Prefer a count: candidate approaches written in twenty minutes; dated partial results; closing-note entries under “one thing that worked”. If the outcome needs judgement, write the material without the condition label, and score it at the end of the test in shuffled order.
4. **Randomise the schedule in advance, in balanced blocks.** Within every block of two or four units, half get each condition, in an order decided by chance. Balancing within blocks stops a slow drift, such as the problem getting easier, from posing as an effect. The schedule script does this. Write the schedule down before the first unit, and do not swap days.
5. **Enough units.** For daily moves, aim for at least eight units of each condition. For regimes, accept that two or three blocks of each is weak evidence.
6. **Decide in advance what would change your practice,** for example “walk days produce at least one more candidate per session”.
7. **Analyse.** The analysis script reports the difference between conditions, and a randomisation test that reshuffles the labels within blocks. The test is valid because the assignment was random. Read the size of the difference first and the p -value second.

The limits are plain:

- You cannot be blind to which condition you are in, and expectation can move the outcome. Counts reduce this but do not remove it.
- Adjacent days can carry over into each other.
- A result about you says nothing about anyone else.

The protocol is still a large improvement on remembering how the month felt.

Part IV. What the evidence supports, and where the accounts disagree

The table sets out, for each practice, the best evidence available, its size where a size exists, and the qualification a fair critic would insist on. Effect sizes are Cohen's d unless stated, and sample sizes are given so that you can weigh each claim rather than take it.

Practice	Best evidence	Size	Qualification
Put it down and return (M11)	Sio and Ormerod (2009): 117 comparisons, 3,606 participants. Liu et al. (2026): 52 studies, 195 effects. McDaniel et al. (2025): preregistered, 200	$d = 0.29$ [0.21, 0.39]; $d = 0.39$ [0.25, 0.53]; no general benefit of a break	Median study size 25. The preparation effect holds only for some problem types. Bias correction for the 2026 estimate not known; a neighbouring literature collapsed under correction
Choice of filler (M14)	Sio and Ormerod subgroups; McDaniel et al. (2025), preregistered, 200 participants; Baird et al. (2012), 145; Murray et al. (2024), 443	0.52 / 0.46 / 0.24, differences not significant; no difference between fillers; about 40% (read from a graph), seen problems only; no support	Weak and inconsistent; no rule follows
Sleep on it (M12)	Wagner et al. (2004): 3×22 trained, 2×20 untrained; later tests (Craig 2018; Brodt 2018, 62; Schönauer 2018, 77; Vickrey and Lerner 2023, 72)	59% vs 23%; waking rest doubled insight; sleep added nothing; nulls; 18% vs 0%, $p = .09$ one-tailed	One strong result with weak support since. Task decides (Lerner and Gluck, 2019). The nap-stage studies conflict and did not randomise
Walk (M8, M13)	Thabane et al. (2026): 23 studies, 1,036 participants; Oppezzo and Schwartz (2014)	Divergent $d = 0.93$ [0.44, 1.42]; convergent $d = 0.16$ [-0.31, 0.63]	Much of it from one laboratory. No evidence of harm to single-answer work
Simplest example (M1)	Wilkerson-Jerde and Wilensky (2011), 10; Lockwood et al. (2016), 220 answers and 19 interviews	Descriptive only	A reading task and set conjectures. About half the survey respondents were doctoral students. Shows that experts do it, not that it helps
Counterexample hunt (M2)	Savic (2015), 9; Lockwood et al. (2016)	3 of 6 tried counterexamples; none recovered that way	A set task in prepared notes; tiny sample; no comparison condition
Switch to other mathematics (M14, M15)	Savic (2015); Sriraman (2004), 5; Johansen and Misfeldt (2014), 13	6 of 6 moved on in the notes	The notes supplied the next item. Sriraman's own excerpts include exceptions. Wiles dissents
Talk before ready (M10)	Cranshaw and Kittur (2011): one project, 1,228 numbered comments, 39 commenters	22 of 39 (56%) wrote a milestone comment; 5 light commenters wrote 6	One project. Milestones chosen by the participants. The two leaders wrote 42%

Practice	Best evidence	Size	Qualification
Change representation (M5)	Tabatabaeian et al. (2025): 6 mathematicians, 12 sessions, 24 insights	Surprisal 1.23 before insight vs 1.07 otherwise, $P < 0.01$	Competition problems. Correlational. Surprisal was also high just after insight
Accept the state (M20)	Ross and Arfini (2024): 4 preregistered experiments, 856 participants	Motivation did not differ between resolved and unresolved impasses	Laboratory impasses lasting minutes
Machine-assisted work (M23)	Collins et al. (2024); Feng et al. (2026); Bäuerle et al. (2026); Collins et al. (2026)	13 of 200 graded answers meaningfully correct; tools raised correct formalisation by 0.30	Small and recent; two by the tool developers; three not peer reviewed
Hours, portfolio, wastebasket, rising sea, immersion (M15–M19)	Testimony only	None	From exceptionally successful people in unusual circumstances, and contested within the testimony itself. Your own log, run as in Instrument I7, is the only available test

Seven disagreements the compilation does not resolve

1. Immersion or abandonment.

- *For immersion*: Bollobás says bear the problem in mind all the time. Hamming says saturate yourself so that the unconscious has nothing else to work on. Littlewood says a month's immersion makes the mind know its problem as the tongue knows the mouth. Wiles, stuck, can think of nothing else.
- *For abandonment*: Hadamard abandoned problems for a while and told beginners to do the same. Poincaré's illuminations came after travel had made him forget his work, and after he had gone to the seaside and thought of something else.

Both camps appeal to unconscious work. The laboratory cannot reach the timescale, though Gilhooly et al. (2012) suggest that for tasks asking for ideas, an immediate break can work. M19 asks you to test both on one problem.

2. Hours.

- *Hardy*: an unbroken morning, by Snow's report.
- *Littlewood*: four or five hours with a break every hour, on what his text calls days free from research (probably days without teaching). Either work all out or rest completely.
- *Poincaré*: an hour or two a day at his work table, in the period that led to the Fuchsian functions.
- *Huh*: about three hours of focused work, a budget that includes preparing lectures and errands.
- *Connes*: mathematicians work most intensely lying in the dark on a sofa, which fits no schedule at all.

3. Doors. Hamming:

"He who works with the door open gets all kinds of interruptions, but he also occasionally gets clues as to what the world is and what might be important."

— Richard Hamming, "You and your research" (1986)

McDuff says interaction is vital while one is learning, and work by oneself once one has the context. Uhlenbeck thinks alone and finds others necessary. Hardy's mornings, as Snow describes them, were his own.

4. Force or wait. Grothendieck names the chisel and the rising sea, and sides with the sea. Poincaré's one recorded night of forcing, the black coffee, worked, and he calls it contrary to his custom. Tao's "patient, thoughtful attack" with few Eureka moments is a third thing: continuous moderate pressure with the failures recorded.

5. Keep the record, or forget it.

- *Keep it:* Tao writes everything down and keeps even wrong work, because something may be salvaged. Halmos and Littlewood each give rules for ending a session well.
- *Forget it:* Wiles thinks too good a memory is a handicap, because remembering every failed attempt stops you trying the near-variant that works. Uhlenbeck does not save her notes and starts each problem fresh.

M17 suggests recording the failing step rather than the verdict, but that is the compiler's proposal, not a resolution.

6. Portfolio or single problem.

- *Portfolio:* Sarnak, Rota (after Feynman), Hamming and Antieau, together with several of Johansen and Misfeldt's respondents. Villani's version is sequential: move on to the next problem.
- *Single problem:* Wiles cannot work that way, and one of Sriraman's five says he must think on one thing and not switch so much.

7. Where the work happens. The retrospective accounts point to the walk, the shower and waking from sleep, and to moments such as Poincaré's bus stop and Duminil-Copin's swim. The one study that recorded mathematicians at an impasse as it happened saw them moving on to the next theorem in their notes; the notes made that easy. The memorable and the typical are different, and a workbook built only on the memorable would teach the wrong thing.

What is missing from the evidence

- **Emotions.** There is no study of research mathematicians' emotions across a months-long impasse, beyond interview remarks such as those collected by Johansen and Misfeldt (2014). The impasse-emotion literature concerns students and laboratory problems.
- **Interventions.** There is no intervention study of walking, sleep or scheduling in research mathematicians.
- **Diaries.** There is no study based on research diaries.
- **Tasks.** The studies that recorded mathematicians at an impasse all used tasks set by the researchers: notes on semigroups, Putnam problems, a paper to read. Polymath and the machine-assisted studies concern real research, but none of them recorded impasses as such.
- **Bias correction.** An updated incubation meta-analysis now exists (Liu et al., 2026), but whether its estimate survives correction for publication bias is not known from the abstract.
- **Machine-assisted work.** The studies are a year or two old and small, and two of them are by the tool developers.

Every claim in this workbook stands or falls on your own record. That is why Part III is longer than it looks, and why Instrument I7 is there.

Part V. Prior art, and what this workbook adds

The question that prompted this compilation was whether mathematicians' accounts of impasse exist and whether they could be compiled into a workbook. The accounts exist in abundance. As for the compilation, a search in September and October 2026 found nothing that does what this workbook tries to do, though several works come close from different directions.

The three closest books.

- **Mason, Burton and Stacey, *Thinking Mathematically* (1982; 2nd edn 2010)**, is closest in form. It has two chapters on the stuck state ("Responses to being STUCK" and "Still STUCK?") and asks readers to keep a written record of their work. Its problems are self-contained puzzles and its authority is pedagogical; it draws on no research mathematician's account of research.
- **Borwein, Liljedahl and Zhai, *Mathematicians on Creativity* (MAA, 2014)**, is closest in source material.
 - *Structure*: 21 chapters, lettered by the surname of the mathematician quoted, with two appendices.
 - *Contemporary material*: answers from prize-winning mathematicians to seven questions, five of them taken from Hadamard.
 - *Overlap with Liljedahl (2013)*: the editors, the questions and the respondents overlap, and the two appear to draw on one survey. That is this compilation's inference; if it is right, they are not independent evidence.
 - *What it lacks*: exercises. It is organised by person rather than by the reader's situation, and framed around creativity rather than impasse.
- **The five "Advice to a Young Mathematician" essays in *The Princeton Companion to Mathematics* (2008)** are closest in register: research-level testimony from Atiyah, Bollobás, Connes, McDuff and Sarnak, unstructured and without exercises.

Further neighbours:

- **Leone Burton, *Mathematicians as Enquirers* (2004)**: an interview study of research mathematicians, with chapters on intuition and on choosing problems.
- **The Tricky (started by Gowers and others; no longer updated)**: a wiki of research-level problem-solving techniques entered through the question of what kind of problem one is trying to solve. It is organised by the solver's situation, as this workbook is, but it does not draw on accounts of impasse.
- **Johansen and Misfeldt (2014)**: a Danish interview study of thirteen research mathematicians, with a section on frustration and waiting.
- **Schoenfeld's problem-solving course (described in Schoenfeld, 1992)**: trains responses to being stuck through the questions on the card in I2, at student level, with before-and-after protocols.
- **John Mason, "On being stuck on a mathematical problem" (2015)**: an essay on the experience of being stuck, written with teachers in mind.
- **Carlson and Bloom (2005)**: an empirical framework for how mathematicians cycle through phases while solving problems, which would add to the observed evidence.
- **The Education Development Center's *Making Mathematics handbook* (2003)**: prompts for school students stuck on research projects of their own, grouped by type of difficulty.
- ***Living Proof* (AMS/MAA, 2019)**: 41 essays on resilience, but its coping strategies concern staying in the profession rather than working a stuck problem.
- **Tao's career-advice pages**: the richest single corpus of one mathematician's heuristics for impasse.

Courses.

- **Courses on doing research.** Examples are Cornell's Math 711, "A Beginner's Guide to Research in Mathematics" (2003), and Hamming's "Learning to Learn" course at the Naval Postgraduate School.
- **Schoenfeld's course** trained students to monitor and abandon failing approaches.
- **On impasse.** This compilation found no course for research mathematicians specifically on practice at an impasse.

What this workbook adds is the intersection of four things:

- research-level accounts, attributed and machine-checked;
- organisation by the reader's state rather than by theme or person;
- exercises done on the reader's live problem rather than on puzzles;
- the empirical evidence graded alongside, with its limits stated and a protocol for testing each move on oneself.

What it lacks is new interviews asking the questions about impasse that neither Hadamard nor Liljedahl asked: not "describe an illumination" but "what do you physically do in the third month?", "how do you decide to abandon?", and "what does your notebook look like on a bad day?". The published accounts are enough to build the book. They are not enough to know whether the book works.

Appendix A. Sources

These are grouped by theme, in APA 7th-edition style. Every source quoted in the workbook was saved and matched as described in Appendix B. Sources cited only for figures were read in full text unless marked “abstract only” or “read through a text-extraction service”. Retrieval date for all online sources: 9 October 2026.

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Rota, G.-C. (1997). Ten lessons I wish I had been taught. *Notices of the American Mathematical Society*, 44(1), 22–25. <https://sites.math.washington.edu/~bviray/links/comm-rota.pdf>

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Appendix B. Verification record

This appendix is generated by the build script from the output of the verification script, so it describes the quotations as printed in this copy.

Method. Every quotation in the workbook is held in one registry (`quotes.yaml` in the bundle) and printed from it. A script matches each quotation against a saved text of its source:

- **Segments.** It splits the quotation at each ellipsis and at each square-bracketed insertion.
- **Exact matching.** Each remaining segment must appear in the source exactly, in the order written. Only these differences are normalised away: the forms of quotation marks and dashes, spacing, and words broken across a line end. Case and punctuation are not normalised, so an added capital or a full stop where the source continues fails.
- **Recorded fixes.** Where a saved text has an optical-character-recognition error or a page-layout artefact, the correction is recorded beside the quotation and listed below.
- **Ledger.** The match position and a fingerprint of each source file are written to `verification/quote_ledger.csv`.

Result. 160 quotations; 160 match; 0 fail.

How each source was obtained.

- **Saved text:** 148 quotations from 60 files, downloaded and converted to text with standard tools.
- **Checked against a scan:** 7 quotations from Littlewood’s essay. The machine-read text of the 1978 *Mathematical Intelligencer* reprint was checked by eye against the scanned pages 116–117, a copy of which is in the bundle (`sources/scans/`).
- **Text-extraction service:** 5 quotations from 4 sources that blocked direct download. The wording was obtained through a web-reading service that returns a model’s extraction of the page, asked for exact wording in short pieces. This is weaker than a downloaded copy. The sources are: the Weil letter (Notices of the AMS, 2005); Gowers, “Solving analysis problems on auto”; the Halmos excerpts in Notices of the AMS (2007); and Snow’s foreword to Hardy’s *Apology*, read from a third-party copy of the Cambridge edition.
- **Translation:** one quotation is translated from Danish by the compiler (respondent R10, M15). The Danish original is what was matched.

Figures. A second script checks 52 of the numerical and factual claims in the evidence boxes. For each, the printed wording must occur in this workbook, and the supporting phrases, chosen to include the figure in its context, must occur in the saved source. Derived figures, such as percentages and Baird et al.’s implied p -values, are recomputed. The build stops if any claim fails. The record is `verification/claims_ledger.csv`. This is a check that the figures are in the source, not a check that they were read in the right sense; that was done by hand. The figures for Lockwood et al. (2016) and Wilkerson-Jerde and Wilensky (2011) were read through the text-extraction service, because the publisher’s page (Lockwood) and the author-hosted copy (Wilkerson-Jerde, a file named `PME2008.pdf` whose extracted title is the 2011 article’s) both blocked downloading.

Recorded corrections to saved texts.

- `connes_walk`: a footnote at a page break interrupts the sentence in the extracted text and is removed before matching.
- `halmos_upbeat`: Marginal words from the facing column of the PDF (“Sometimes I”) interrupt the line in the extracted text and are removed before matching.
- `hadamard_awakened`: source reads “uwakened”, matched as “awakened” (OCR error or footnote marker).

- `hadamard_awakened`: source reads “a solution 5 long”, matched as “a solution long” (OCR error or footnote marker).

Other notes.

- *Thurston*. The MathOverflow original reads “perseverence at at the fog”; the Cornell reproduction quoted here corrects both slips.
- *Quanta, Orlin and Plus*. The ellipsis in Huh’s “intention and willpower ... are highly over-rated” is Quanta’s own, and the one in Wiles’s Heidelberg remarks is Orlin’s. The square brackets in Wiles’s Plus interview are the magazine’s.
- *Snow*. Wording obtained through the text-extraction service from a third-party copy; the archive.org scan of the Cambridge edition is lending-only and could not be read.
- *Mason, Burton and Stacey*. The chapter titles were confirmed from the publisher’s contents and Karaali’s review. The keywords of the book’s written-record “rubric” could not be checked, so Instrument I1 presents its margin marks as this workbook’s own.
- *Abstract only*. Several studies cited for figures were read only as abstracts; Appendix A marks them.

Every quotation. ID is the registry key, which names the speaker; “Used in” is where the quotation appears; “Source” is the saved file, shortened (full paths, speakers and the quoted text are in `verification/quote_ledger.csv`); “Method” is how the source was obtained.

ID	Used in	Source	Method	Result
<code>sarnak_stuck</code>	Front matter	<code>pcm</code>	text	MATCH
<code>wiles_mansion</code>	Part I	<code>nova</code>	text	MATCH
<code>thurston_fog</code>	Part I	<code>thurston cornell</code>	text	MATCH
<code>bollobas_basket</code>	Part I, M17	<code>pcm</code>	text	MATCH
<code>alon_cloud</code>	Part I	<code>alon2009</code>	text	MATCH
<code>alon_newproblem</code>	Part I	<code>alon2009</code>	text	MATCH
<code>villani_discoveries</code>	Part I	<code>villani gs</code>	text	MATCH
<code>tao_counterexample</code>	Part I, M2	<code>be sceptical of your own work</code>	text	MATCH
<code>halmos_hilbert</code>	M1	<code>macc halmos</code>	text	MATCH
<code>mcduff_simplest</code>	M1	<code>pcm</code>	text	MATCH
<code>vakil_toy</code>	M1	<code>vakil students</code>	text	MATCH
<code>schoenfeld_goose</code>	M1	<code>schoen92</code>	text	MATCH
<code>tao_effortless</code>	M2	<code>be sceptical of your own work</code>	text	MATCH
<code>tao_stronger</code>	M2	<code>be sceptical of your own work</code>	text	MATCH
<code>tao_partial</code>	M3	<code>on the importance of partial progress</code>	text	MATCH
<code>tao_months</code>	M3	<code>be patient</code>	text	MATCH
<code>halmos_months</code>	M3	<code>macc halmos</code>	text	MATCH
<code>halmos_fight</code>	M4	<code>ask yourself dumb questions and answer them</code>	text	MATCH
<code>tao_delete</code>	M4	<code>ask yourself dumb questions and answer them</code>	text	MATCH
<code>tao_stupid</code>	M4	<code>ask yourself dumb questions and answer them</code>	text	MATCH
<code>schoenfeld_semester</code>	M4	<code>schoen92</code>	text	MATCH
<code>mirz_drawing</code>	M5	<code>q mirz</code>	text	MATCH

ID	Used in	Source	Method	Result
vondrak_drawing	M5	q mirz	text	MATCH
figalli_drawing	M5	q figalli	text	MATCH
weil_trilingual	M5	weil notices 2005 webfetch	extraction	MATCH
weil_adulterous	M5	weil notices 2005 webfetch	extraction	MATCH
thurston_listening	M5	thurston arxiv	text	MATCH
gowers_auto	M6	gowers autoanalysis webfetch	extraction	MATCH
gowers_lower	M6	gowers analysis	text	MATCH
gowers_cubic	M7	gowers cubic	text	MATCH
polya_related	M7	polya	text	MATCH
connes_walk	M8	pcm	text	MATCH
connes_memory	M8	pcm	text	MATCH
cepelewicz_moving	M8	q dc	text	MATCH
scholze_capacity	M8	q scholze	text	MATCH
tao_writedown	M9	write down what youve done	text	MATCH
halmos_upbeat	M9	halmos write	text	MATCH
littlewood_middle	M9	intelligencer djvu	scan	MATCH
huh_record	M9	imu huh	text	MATCH
tao_salvage	M9	use the wastebasket	text	MATCH
dc_crappy	M10	q dc	text	MATCH
gowers_stupid	M10	gowers poly	text	MATCH
gowers_resist	M10	gowers poly	text	MATCH
mcduff_communal	M10	pcm	text	MATCH
uhl_alone	M10	sciam	text	MATCH
marsden_colleague	M10	liljedahl	text	MATCH
hadamard_abandon	M11	hadamard	text	MATCH
poincare_step	M11	halsted fos	text	MATCH
poincare_fruitful	M11	halsted fos	text	MATCH
poincare_fruitless	M11	halsted fos	text	MATCH
wiles_relax	M11	plus	text	MATCH
uhl_stuck	M11	sciam	text	MATCH
uhl_fresh	M17	sciam	text	MATCH
hadamard_certain	M12	hadamard	text	MATCH
hadamard_awakened	M12	hadamard	text	MATCH
peskin_asleep	M12	liljedahl	text	MATCH
christodoulou_bed	M12	liljedahl	text	MATCH
poincare_coffee	M12	halsted fos	text	MATCH
poincare_bluff	M13	halsted fos	text	MATCH
littlewood_walking	M13	intelligencer djvu	scan	MATCH
gnedenko_outings	M13	kolmogorov	text	MATCH
fleming_campus	M13	liljedahl	text	MATCH
dc_swim	M13	q dc	text	MATCH
wiles_landscapes	M13	plus	text	MATCH
dc_calm	M14	imu dc	text	MATCH
villani_replying	M14	villani gs	text	MATCH
villani_effort	M14	villani gs	text	MATCH
huh_quiet	M14	q huh	text	MATCH
huh_sleepy	M14	q huh	text	MATCH

ID	Used in	Source	Method	Result
sarnak_range	M15	pcm	text	MATCH
rota_dozen	M15	rota	text	MATCH
hamming_problems	M15	hamming	text	MATCH
tao_lowhanging	M15	dont prematurely obsess on a single big problem or big theory	text	MATCH
villani_next	M15	villani gs	text	MATCH
wiles_cantswitch	M15	plus	text	MATCH
alon_threemonths	M21	alon2009	text	MATCH
snow_hardy	M16	snow foreword webfetch	extraction	MATCH
littlewood_hours	M16	intelligencer djvu	scan	MATCH
littlewood_teaching	M16	intelligencer djvu	scan	MATCH
littlewood_allout	M16	intelligencer djvu	scan	MATCH
poincare_hour	M16	halsted fos	text	MATCH
huh_willpower	M16	q huh	text	MATCH
huh_threehours	M16	q huh	text	MATCH
connes_sofa	M16	pcm	text	MATCH
tao_deadend	M17	use the wastebasket	text	MATCH
tao_embarrassing	M17	use the wastebasket	text	MATCH
rota_mistakes	M17	rota	text	MATCH
wiles_memory	M17	plus	text	MATCH
wiles_littlething	M17	plus	text	MATCH
groth_chisel	M18	mclarty	text	MATCH
groth_sea	M18	mclarty	text	MATCH
tao_outside	M18	dont be afraid to learn things outside your field	text	MATCH
atiyah_tangent	M18	pcm	text	MATCH
scholze_linear	M18	q scholze	text	MATCH
bollobas_newton	M19	pcm	text	MATCH
hamming_immersed	M19	hamming	text	MATCH
littlewood_tongue	M19	intelligencer djvu	scan	MATCH
poincare_illumination	M19	halsted fos	text	MATCH
poincare_travel	M19	halsted fos	text	MATCH
poincare_seaside	M19	halsted fos	text	MATCH
wiles_stressful	M20	orlin	text	MATCH
wiles_process	M20	orlin	text	MATCH
wiles_difficult	M20	orlin	text	MATCH
thurston_muddled	M20	thurston cornell	text	MATCH
mirz_disappointed	M20	q mirz	text	MATCH
mirz_confident	M20	q mirz	text	MATCH
tao_eureka	M20	be patient	text	MATCH
tao_range	M21	continually aim just beyond your current range	text	MATCH
tao_warning	M21	dont prematurely obsess on a single big problem or big theory	text	MATCH
villani_unsure	M21	villani gs	text	MATCH
tao_unpredictable	M21	be flexible	text	MATCH
vakil_three	M22	vakil three	text	MATCH

ID	Used in	Source	Method	Result
vakil_vague	M22	vakil three	text	MATCH
vakil_contrived	M22	vakil three	text	MATCH
vakil_drop	M22	vakil three	text	MATCH
uhl_read	M22	ems uhl	text	MATCH
halmos_book	M22	halmos notices 2007 webfetch	extraction	MATCH
hamming_insight	M23	hamming 1997 art of doing science worrydream	text	MATCH
hamming_volume	M23	hamming 1997 art of doing science worrydream	text	MATCH
hamming_simple	M23	hamming 1997 art of doing science worrydream	text	MATCH
georgiev_inexperience	M23	georgiev gomezterrano tao wagner alphaevolve 2025	text	MATCH
georgiev_cheat	M23	georgiev gomezterrano tao wagner alphaevolve 2025	text	MATCH
georgiev_deep	M23	georgiev gomezterrano tao wagner alphaevolve 2025	text	MATCH
antieau_more	M23	antieau fast math slow math 2026	text	MATCH
bailey_rule_out	M23	bailey borwein experimental mathematics notices 2005	text	MATCH
tao_align	M23	tao story erdos 1026 2025	text	MATCH
tao_code	M23	tao machine assisted proof 2025 nsfpar columns	text	MATCH
williams_change	M23	abouzaid etal first proof 2026	text	MATCH
scholze_ram	M23	scholze half year lte 2021	text	MATCH
scholze_terrain	M23	scholze half year lte 2021	text	MATCH
aaronson_wrong	M23	aaronson qma singularity 2025	text	MATCH
aaronson_sweet	M23	aaronson qma singularity 2025	text	MATCH
tao_o1	M23	tao mastodon o1 2024 09 13	text	MATCH
etp_llm	M23	bolan etal equational theories project 2025	text	MATCH
buzzard_lean	M23	buzzard can ai do maths yet 2024	text	MATCH
collins_lazy	M23	collins etal pnas 2024 pmc	text	MATCH
schoenfeld_hell	M4	schoen92	text	MATCH
schoenfeld_army	M4	schoen92	text	MATCH
savic_some	M11	savic	text	MATCH
savic_almost	M11	savic	text	MATCH
littlewood_relaxed	M13	intelligencer djvu	scan	MATCH
hadamard_dream	M12	hadamard	text	MATCH
r10_nowhere	M15	mona	text	MATCH
poincare_hourortwo	M16	halsted fos	text	MATCH
liljedahl_further	M20	liljedahl	text	MATCH
gowers_gpt4	M23	collins etal pnas 2024 pmc	text	MATCH
feng_arduous	M23	feng erdos semiautonomous 2026	text	MATCH

ID	Used in	Source	Method	Result
bauerle_cheating	M23	bauerle etal intentmaking 2026	text	MATCH
bauerle_loops	M23	bauerle etal intentmaking 2026	text	MATCH
schoenfeld_q1	I2	schoen92	text	MATCH
schoenfeld_q2	I2	schoen92	text	MATCH
schoenfeld_q3	I2	schoen92	text	MATCH
schoenfeld_hellphrase	I2	schoen92	text	MATCH
hamming_door	Part IV	hamming	text	MATCH
tao_patient	Part IV	be patient	text	MATCH
antieau_rule	M23	antieau fast math slow math 2026	text	MATCH
antieau_burners	M15	antieau fast math slow math 2026	text	MATCH
halmos_tomorrow	M9	halmos write	text	MATCH
thurston_derivative	M5	thurston arxiv	text	MATCH
thurston_continues	M5	thurston arxiv	text	MATCH